# Monetary policy with and without time to build

October 2026

## Model

Wage fixed: $W_t = \bar{W}$.

**Euler equation**

$$\Xi_t \frac{C_t^{-\gamma}}{P_t} = \beta(1 + i_t)\, \Xi_{t+1} \frac{C_{t+1}^{-\gamma}}{P_{t+1}}$$

**Price index**

$$P_t = \left( \sum_j \Big[ \frac{\bar{W} \prod_{s=t-\tau_j}^{t-1} (1 + i_s)}{\zeta_j \, A_{t-\tau_j}} \Big]^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$$

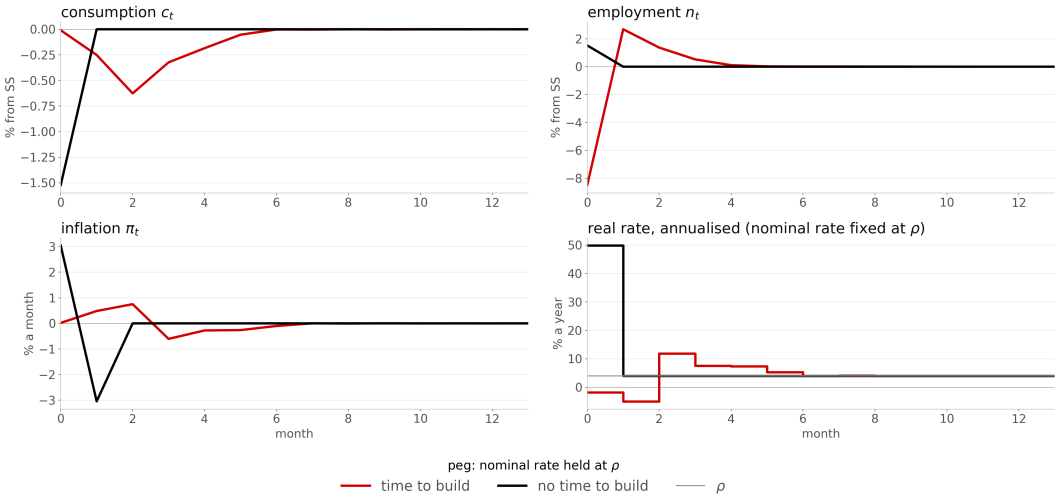
## Planner's problem

$$\max_{\{i_t\}} U = \sum_t \beta^t \Xi_t \Big[\frac{C_t^{1-\gamma}}{1-\gamma} - \chi \frac{L_t^{1+\eta}}{1+\eta}\Big]$$

s.t. Euler, prices, price index, $C_t = Y_t$; $\quad i_t \in [0, 5\%]$ a month

- $L_t$: labour **hired** at $t$ (for goods delivered at $t + \tau_j$)
- One period = one month; rate free from the news date (or $t = 0$) to $t = 7$, then $\rho$
- Shocks: TFP $A_0 = 0.97$; $\quad$ demand $\Xi_0 = e^{-0.01}$
- $\gamma = \eta = 2,\ \chi = 1,\ \sigma = 5,\ \kappa = 0.938$; $\rho = 0.33\%$ a month ($4\%$ a year); price = marginal cost

# Expected shocks, policy from $t = 0$: TFP shock, no policy change



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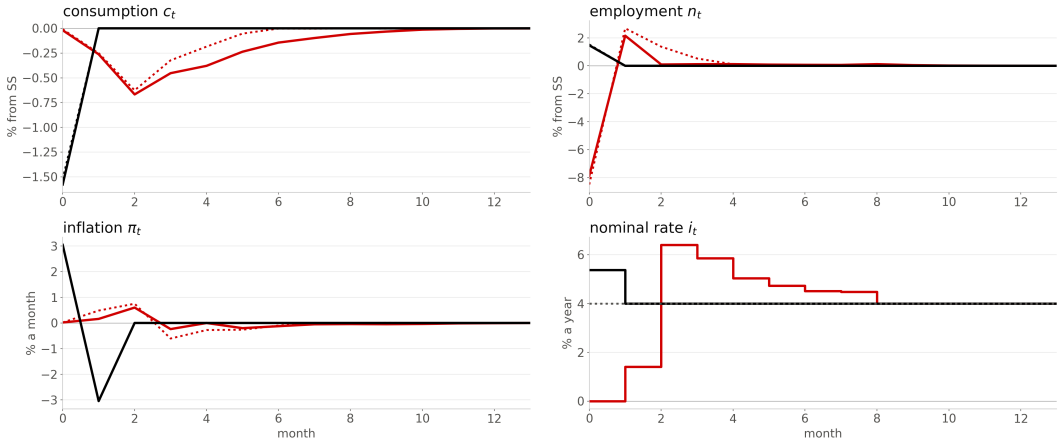
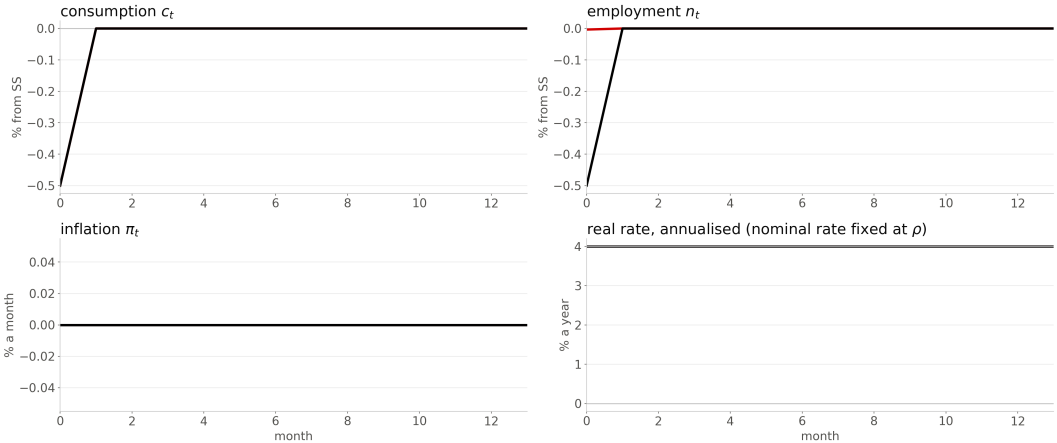
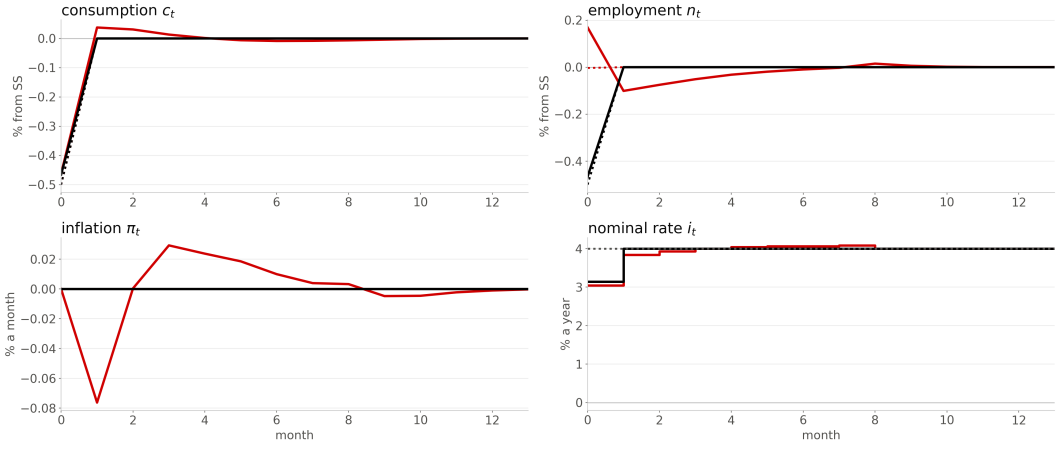


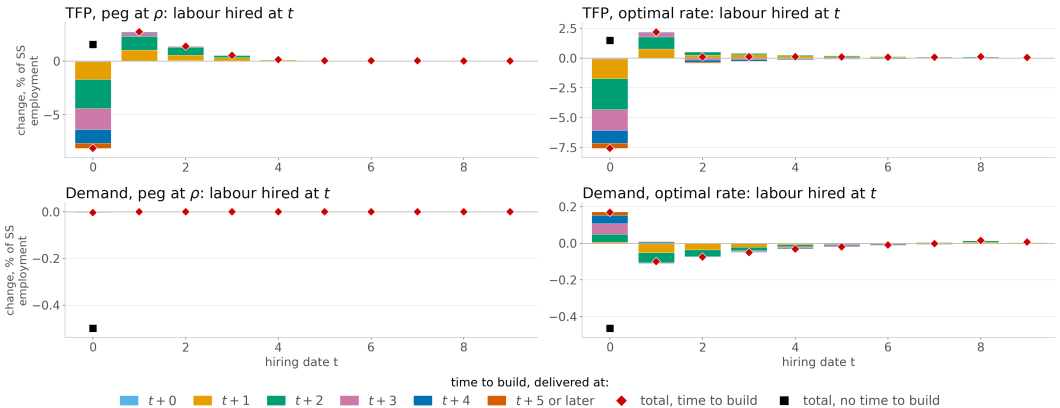
# Expected shocks, policy from $t = 0$: demand shock, no policy change



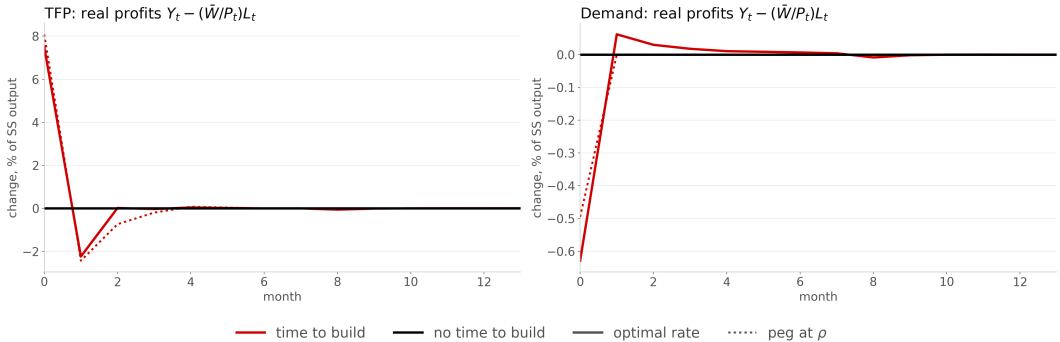
# Expected shocks, policy from $t = 0$: demand shock



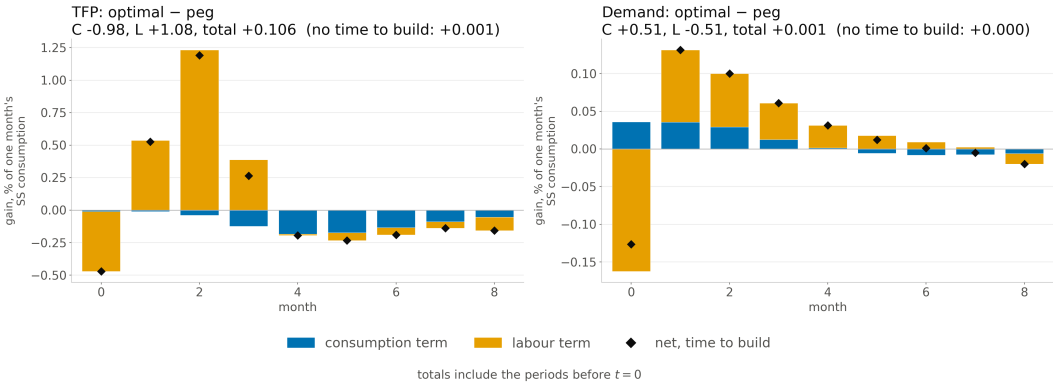
# Expected shocks, policy from $t = 0$: where labour goes



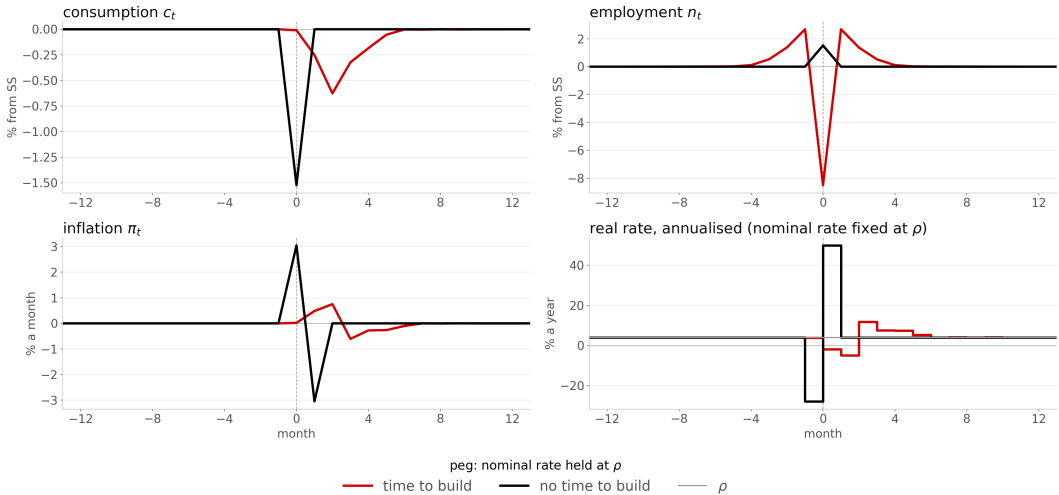
# Expected shocks, policy from $t = 0$: firm profits



# Expected shocks, policy from $t = 0$: welfare gain from optimal policy



# Expected shocks, policy from the news: TFP shock, no policy change



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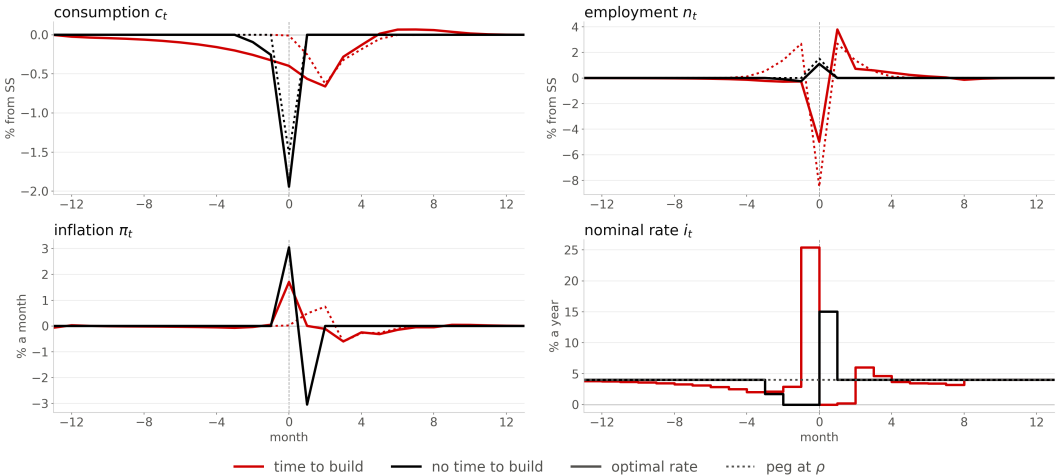
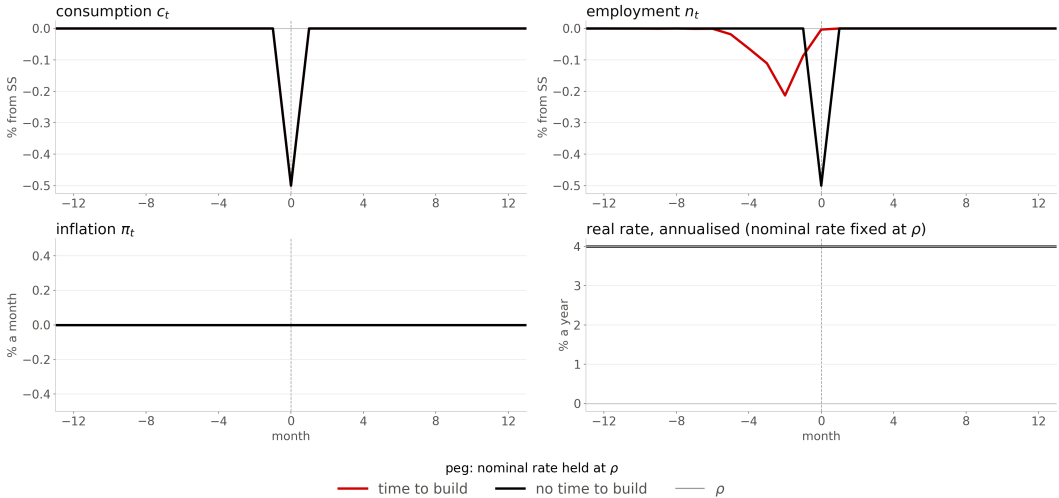
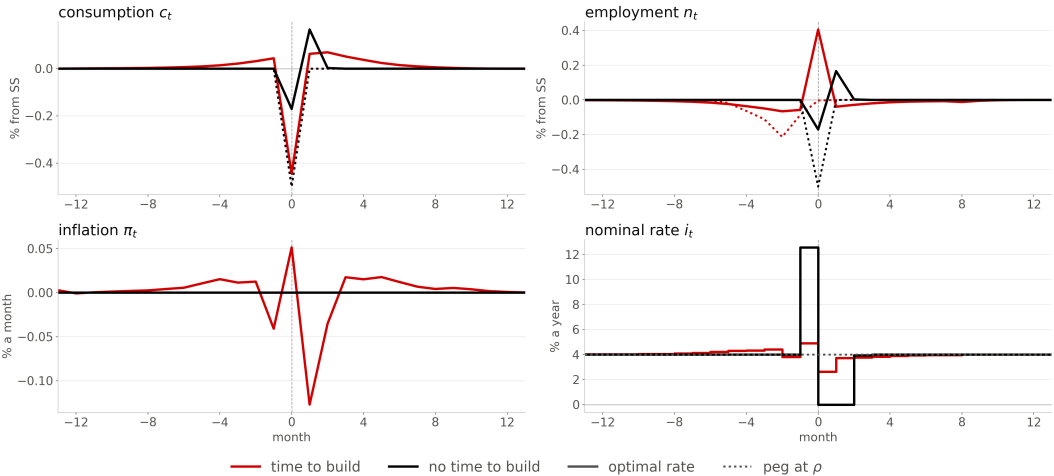


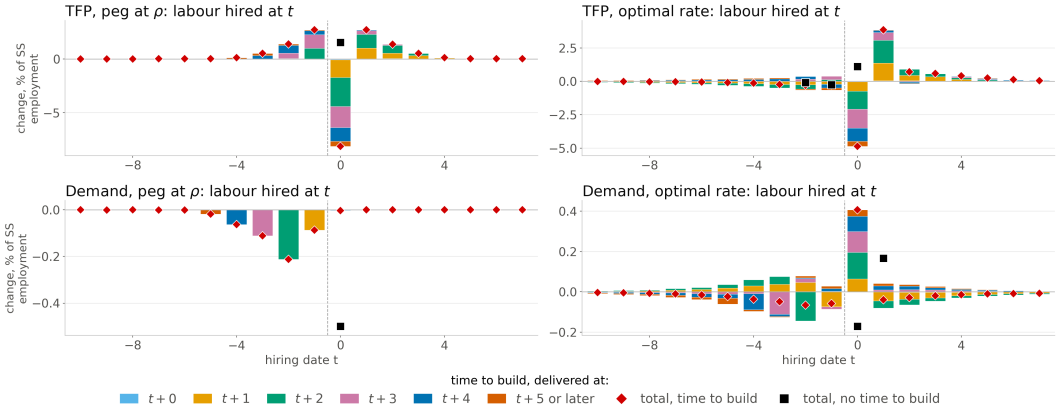
# Expected shocks, policy from the news: demand shock, no policy change



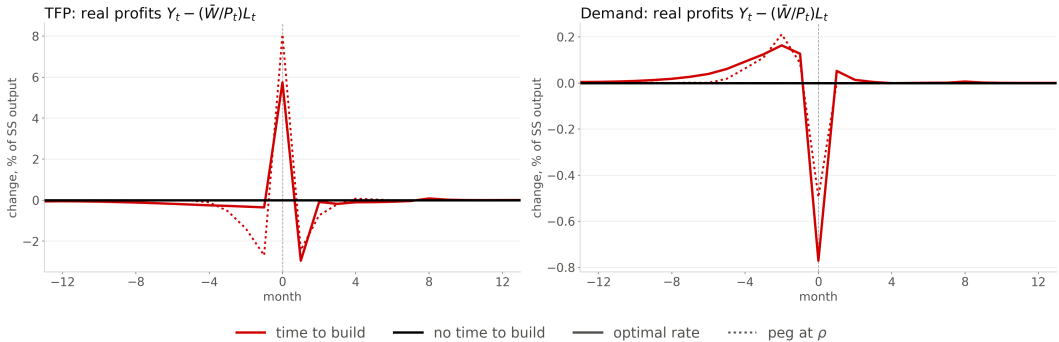
# Expected shocks, policy from the news: demand shock



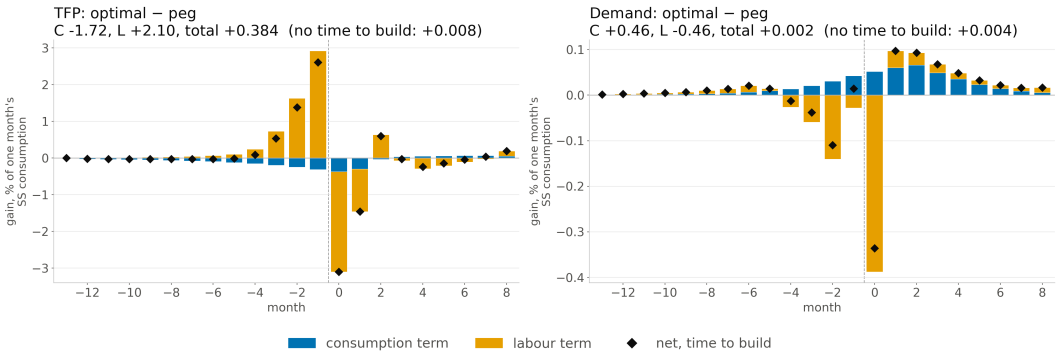
# Expected shocks, policy from the news: where labour goes



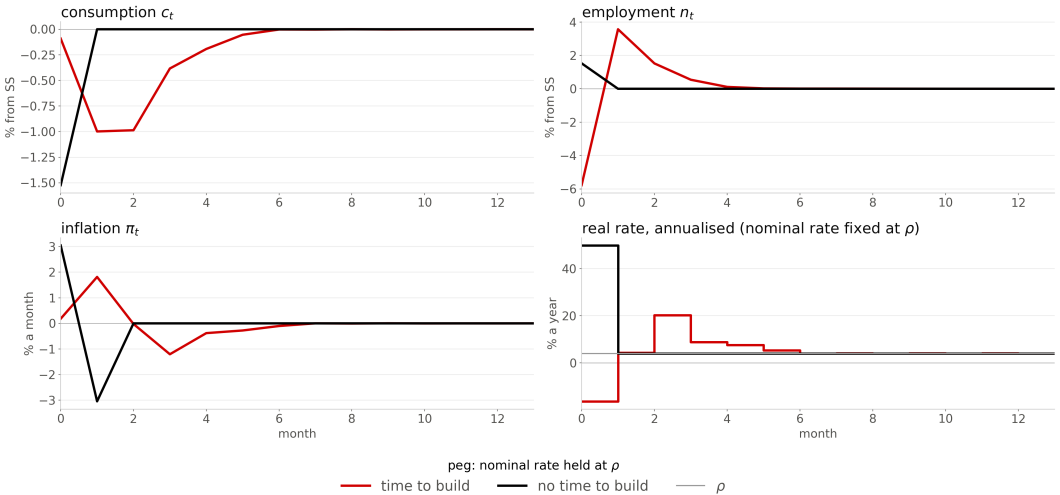
# Expected shocks, policy from the news: firm profits



# Expected shocks, policy from the news: welfare gain from optimal policy



# Unexpected shocks, fixed wage: TFP shock, no policy change



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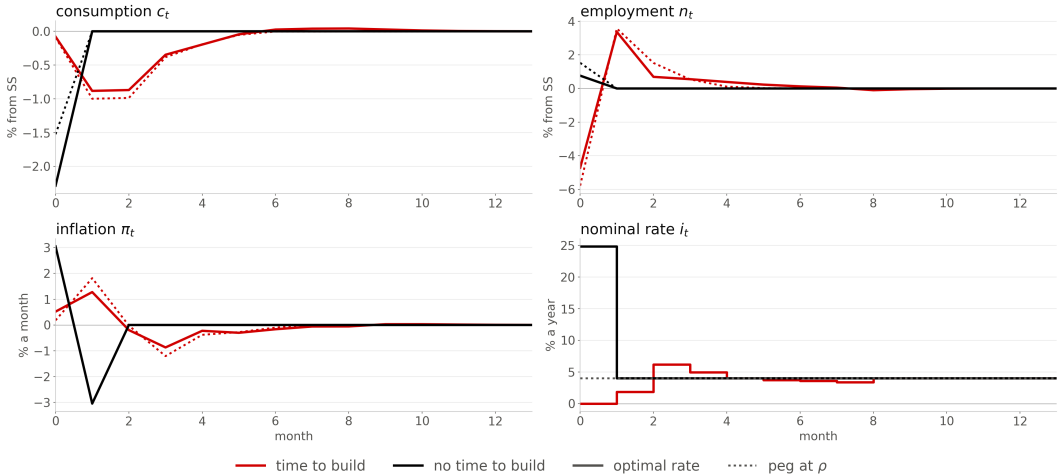
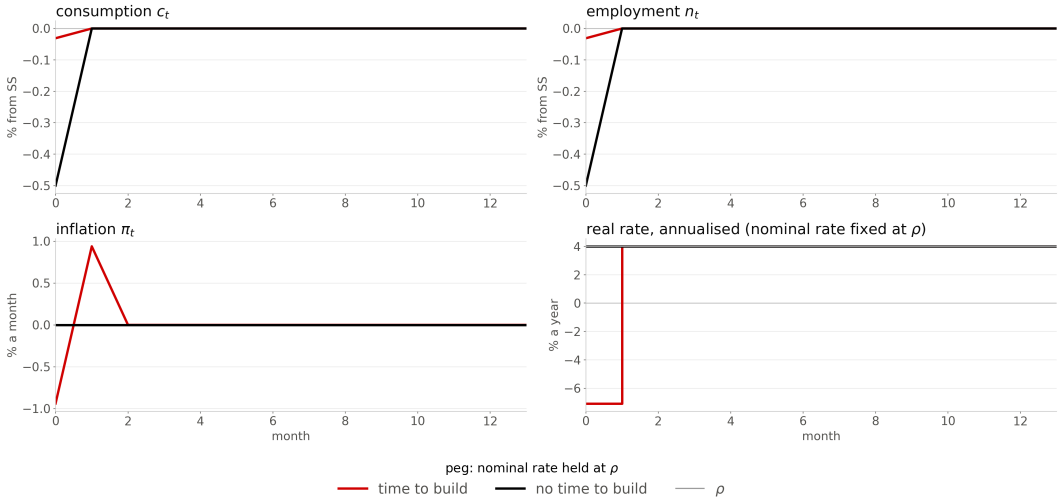


# Unexpected shocks, fixed wage: demand shock, no policy change



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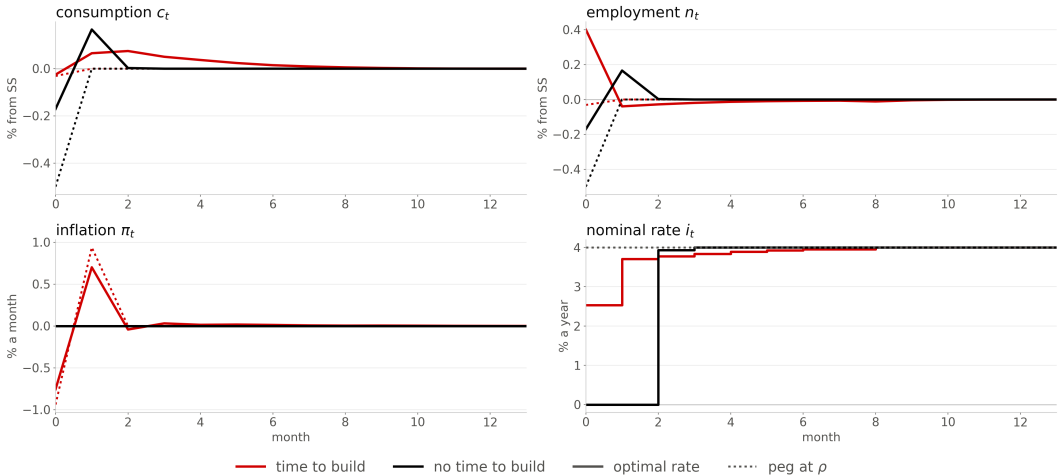
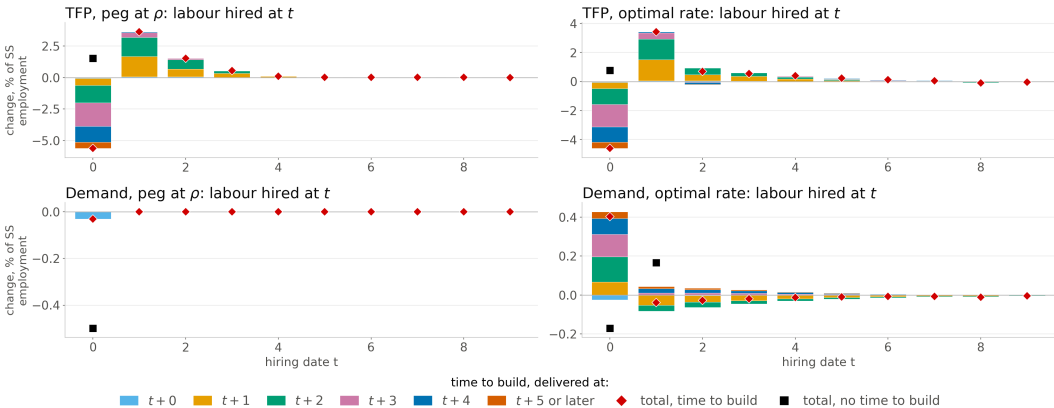
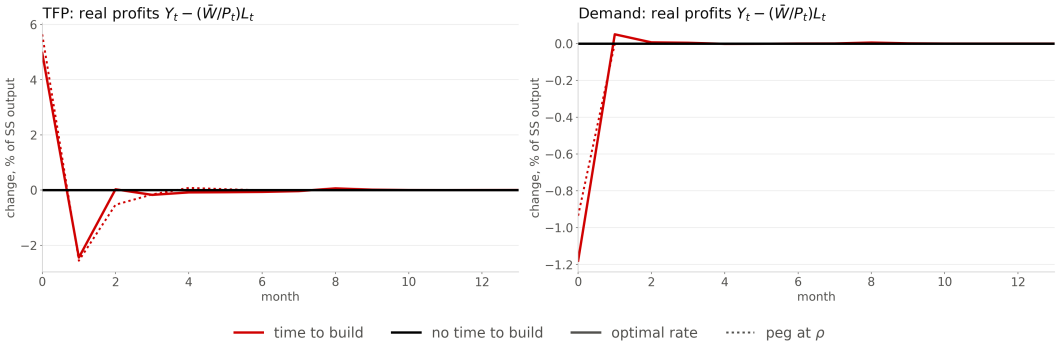


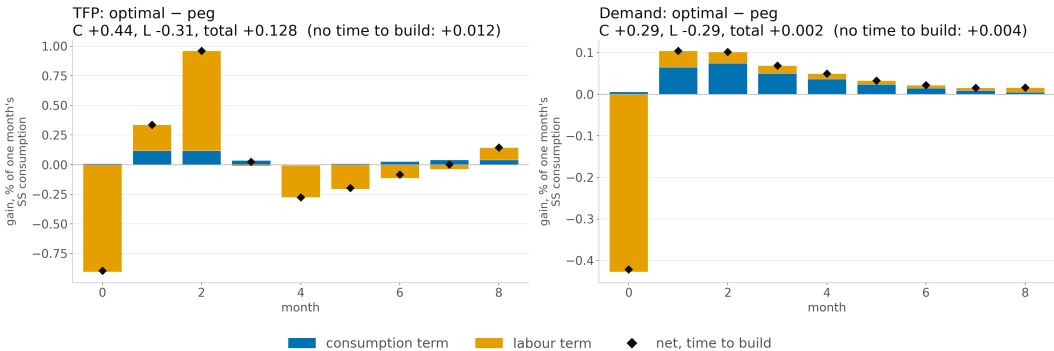
# Unexpected shocks, fixed wage: where labour goes



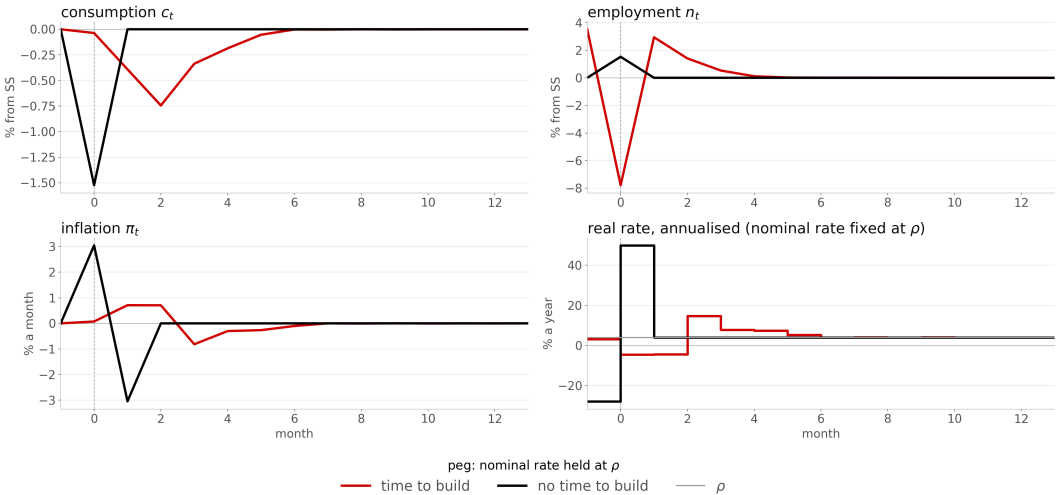
# Unexpected shocks, fixed wage: firm profits



# Unexpected shocks, fixed wage: welfare gain from optimal policy



# Shock announced 1 period ahead: TFP shock, no policy change



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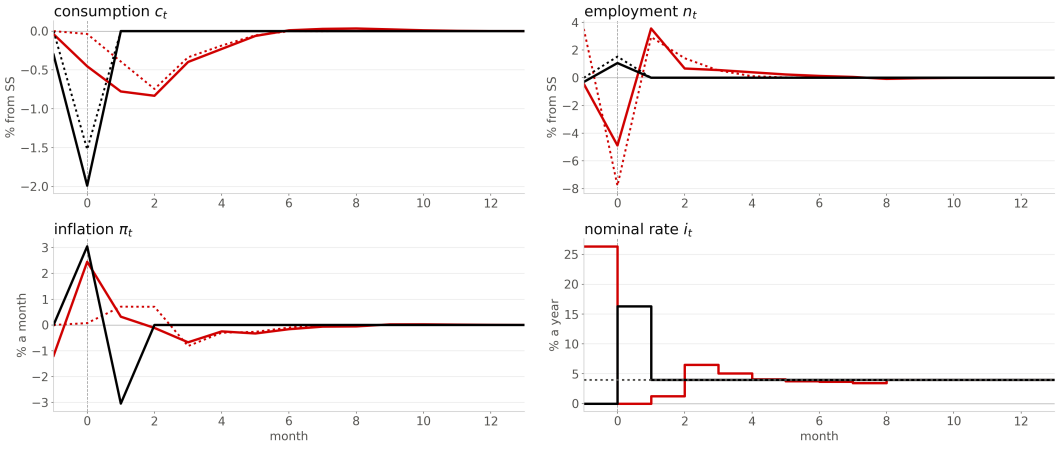
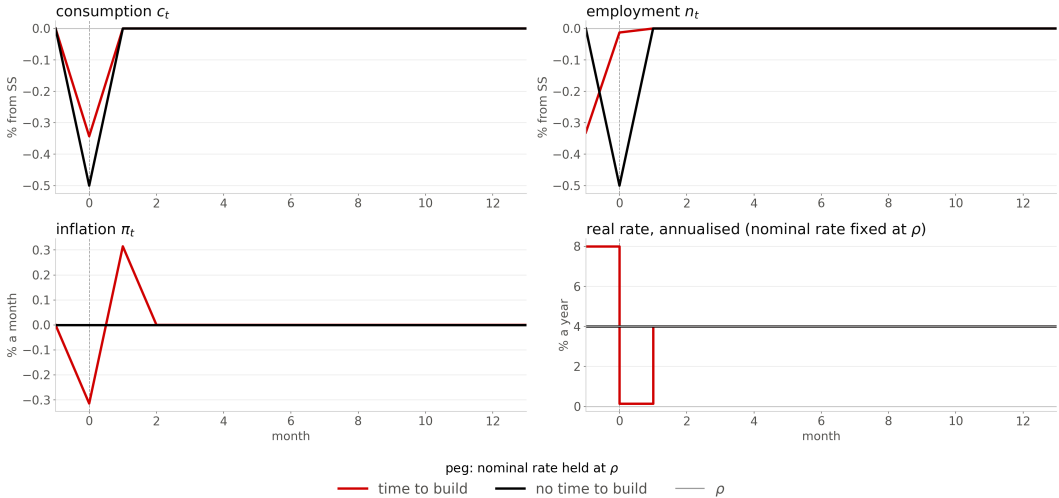


# Shock announced 1 period ahead: demand shock, no policy change



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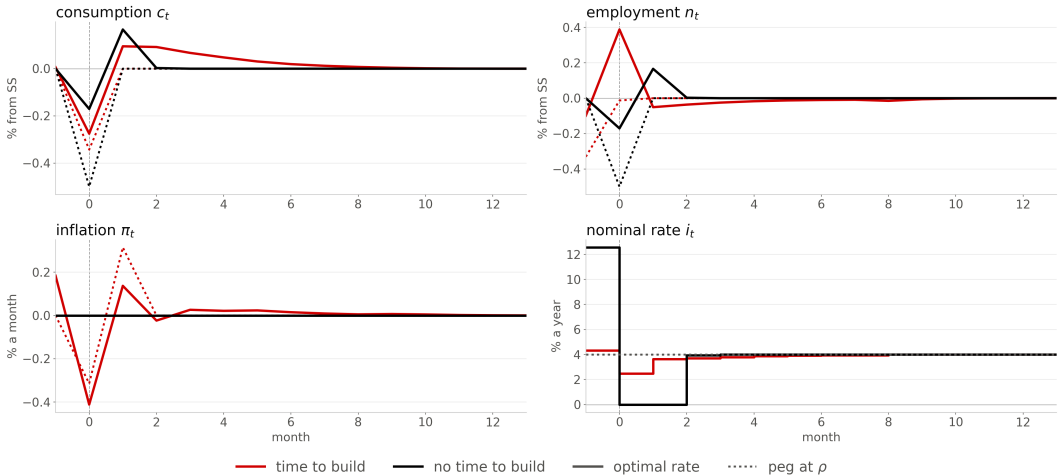
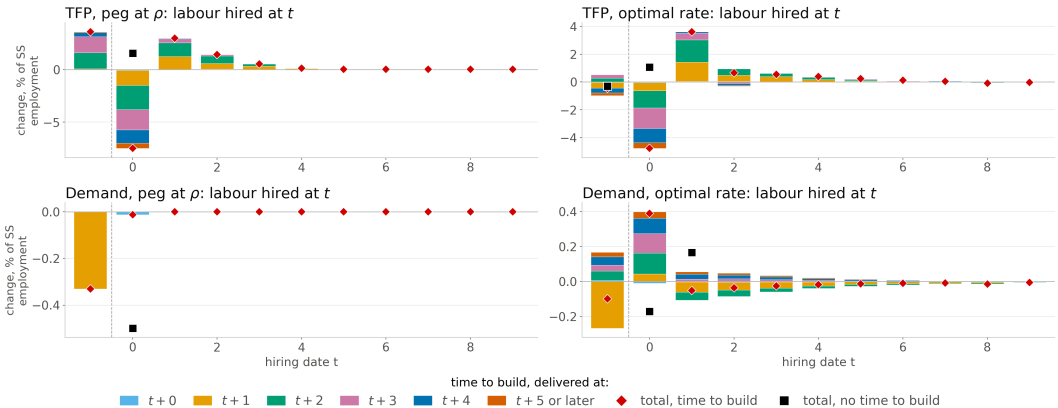
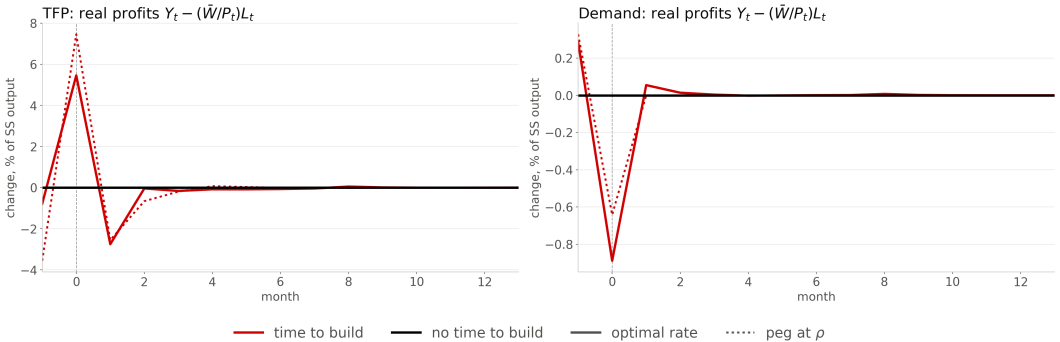


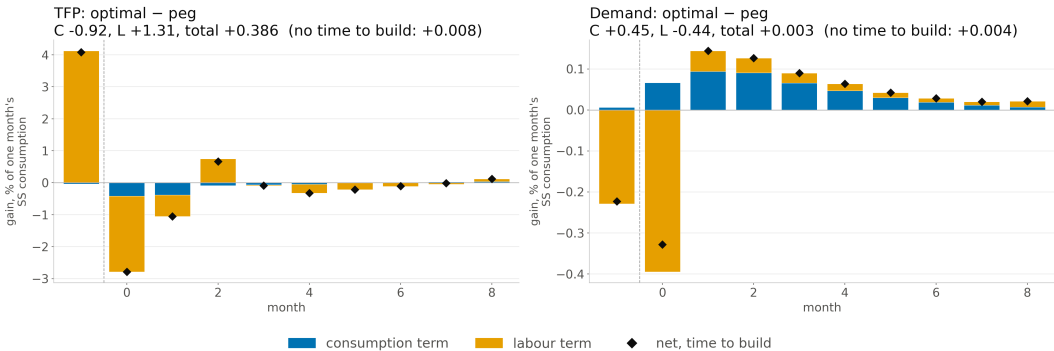
# Shock announced 1 period ahead: where labour goes



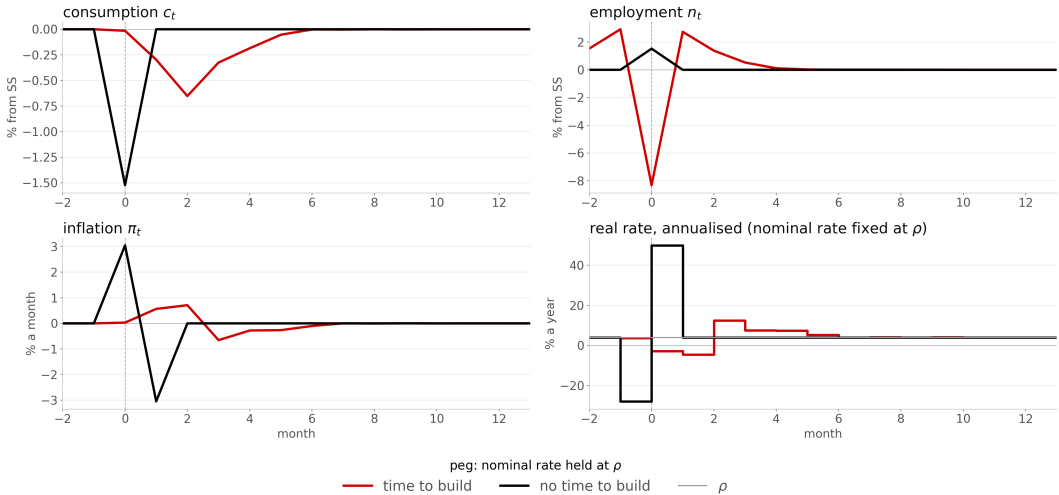
# Shock announced 1 period ahead: firm profits



# Shock announced 1 period ahead: welfare gain from optimal policy



# Shock announced 2 periods ahead: TFP shock, no policy change



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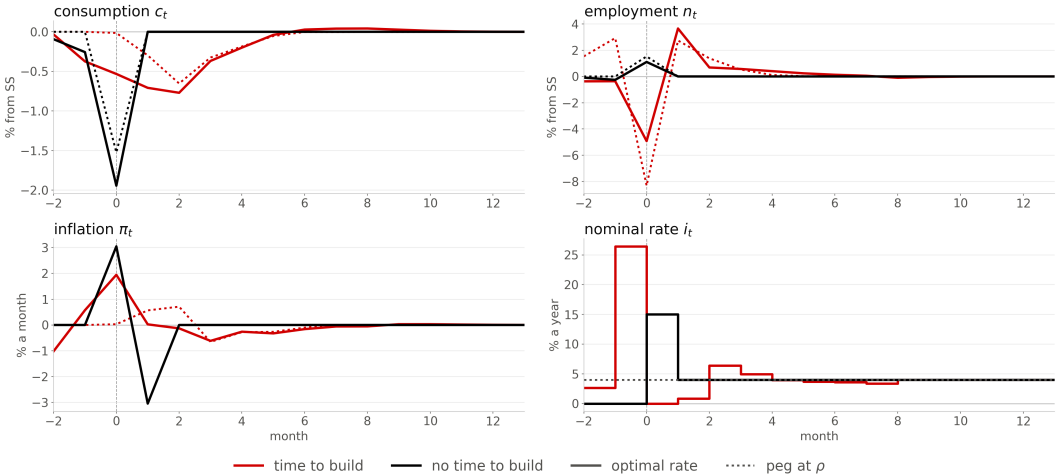

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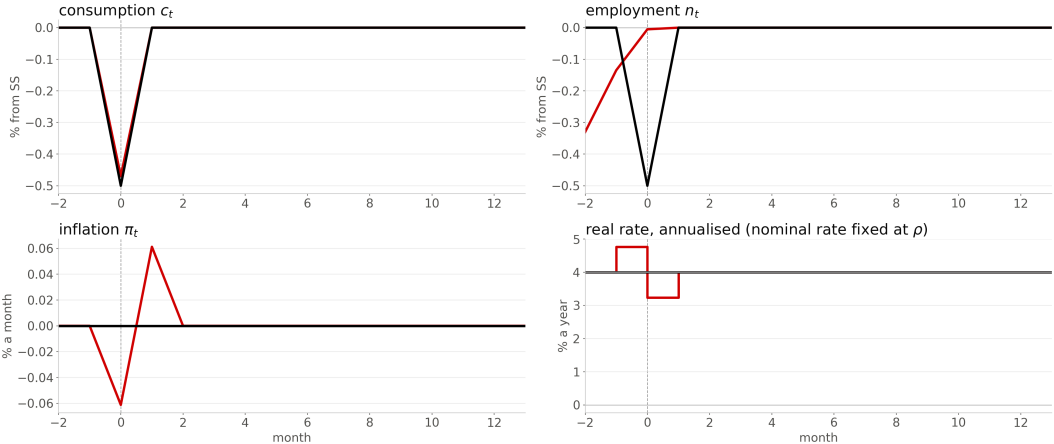

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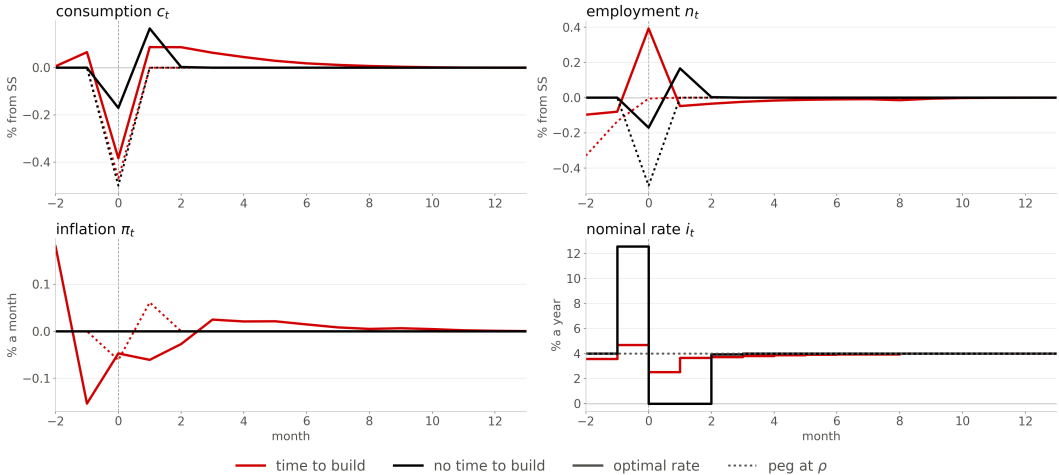
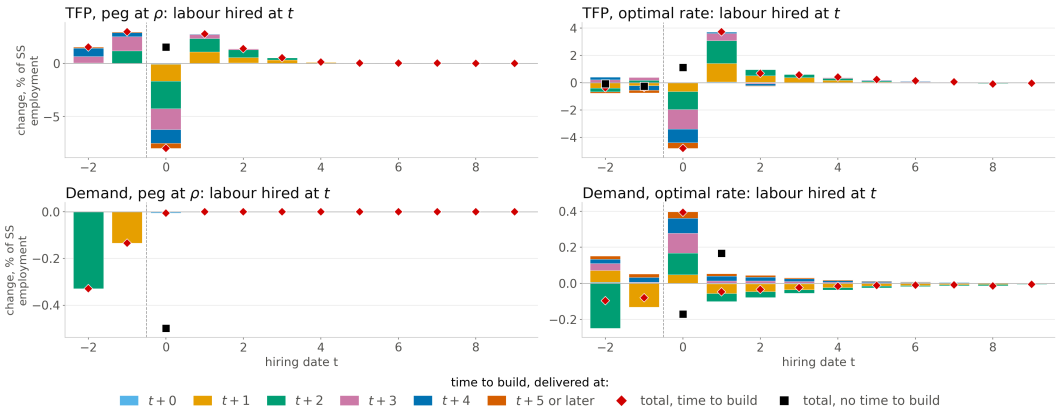
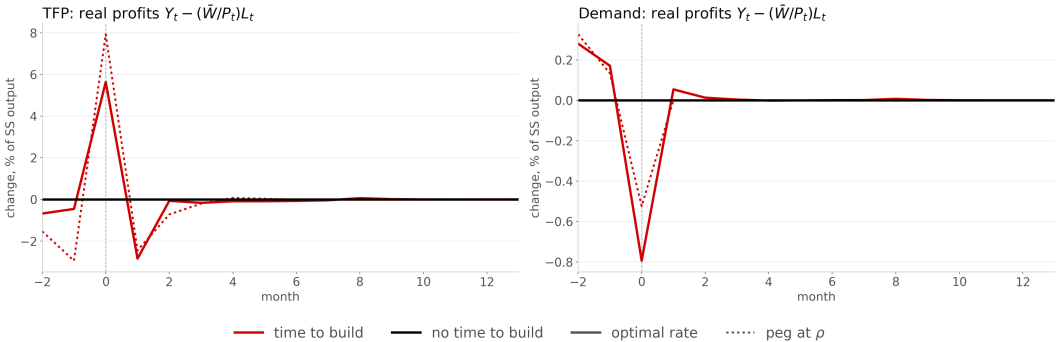


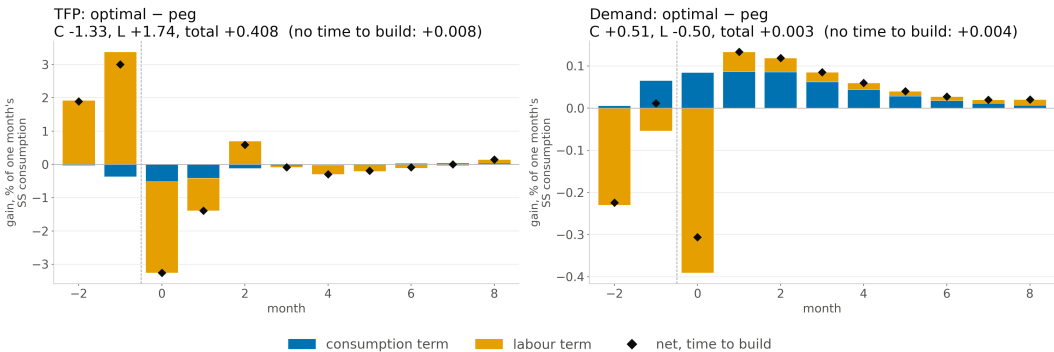
# Shock announced 2 periods ahead: where labour goes



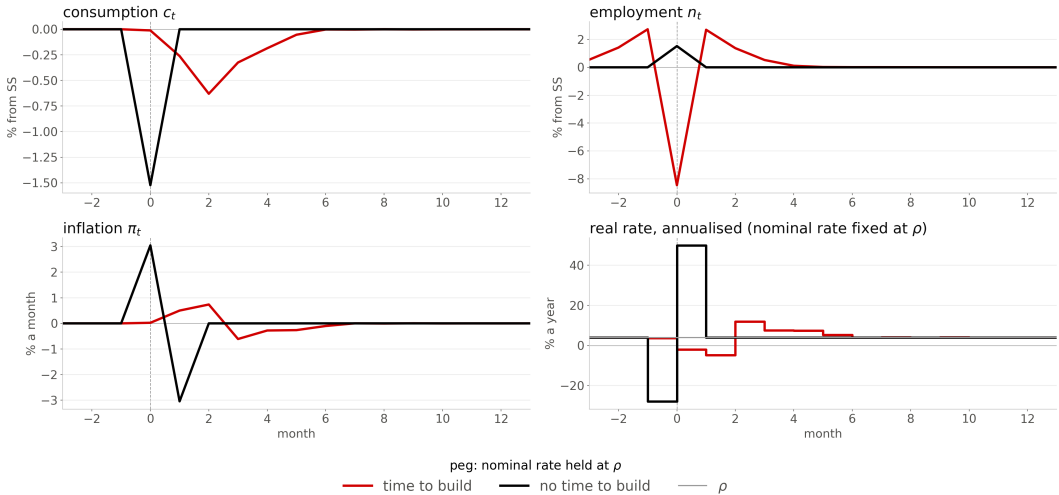
# Shock announced 2 periods ahead: firm profits



# Shock announced 2 periods ahead: welfare gain from optimal policy



# Shock announced 3 periods ahead: TFP shock, no policy change



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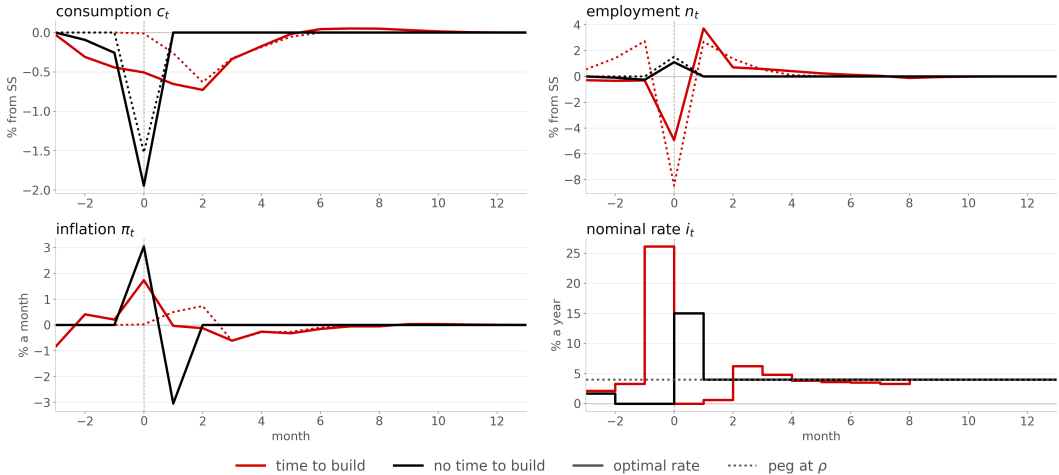

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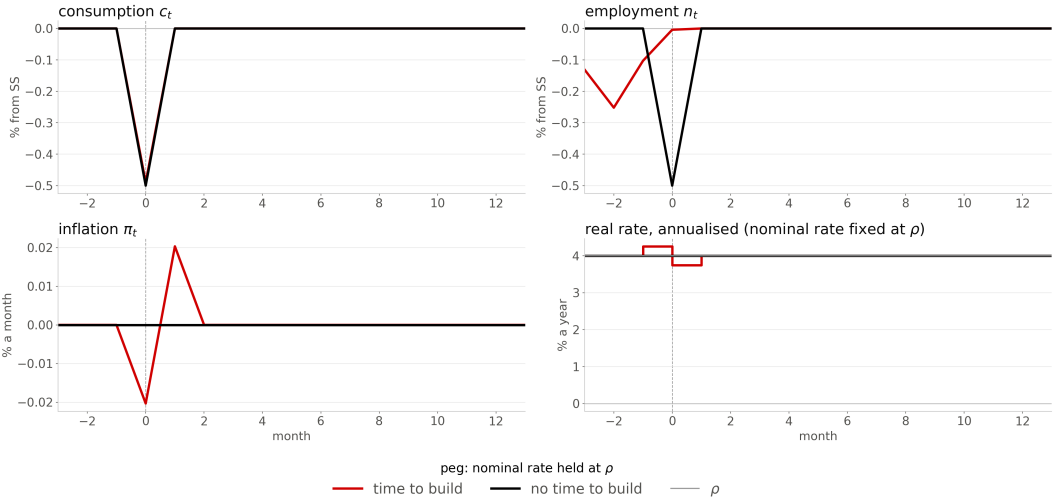

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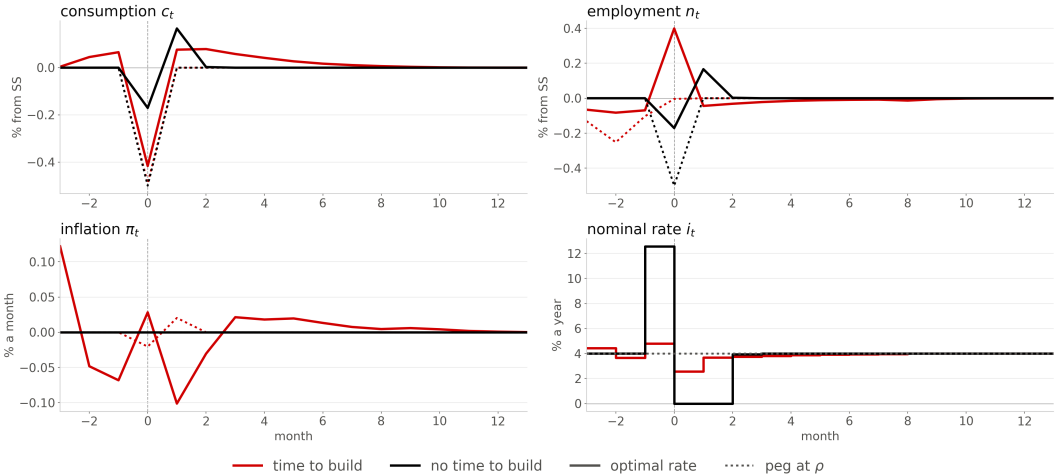
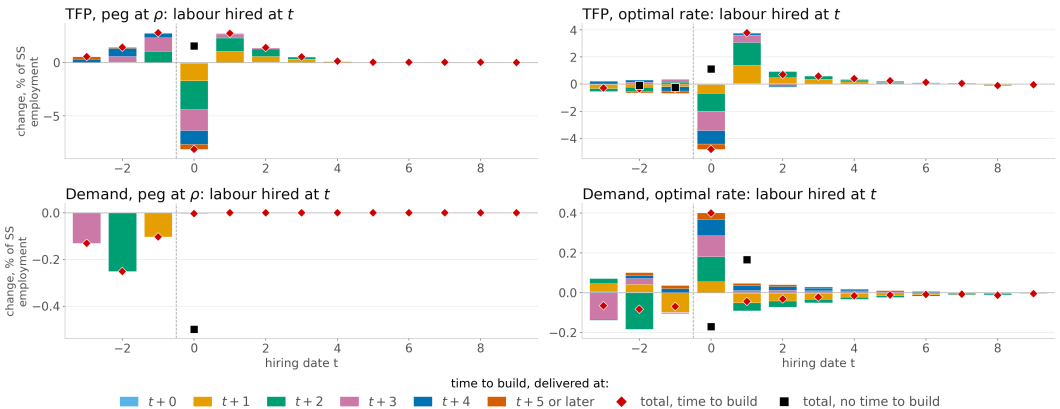
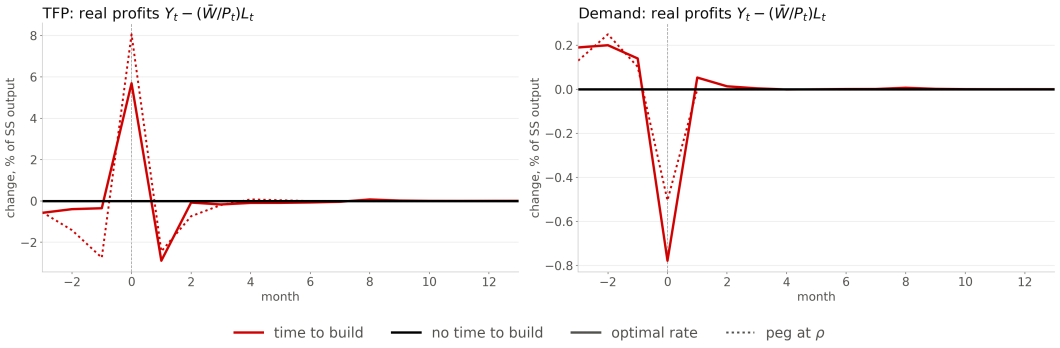


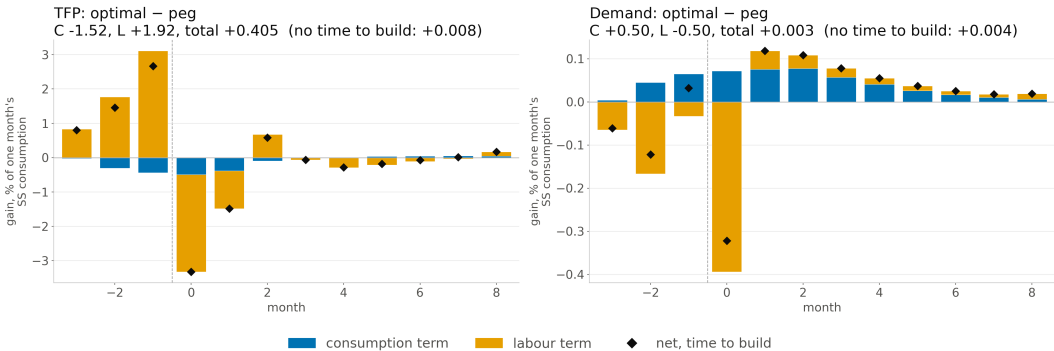
# Shock announced 3 periods ahead: where labour goes



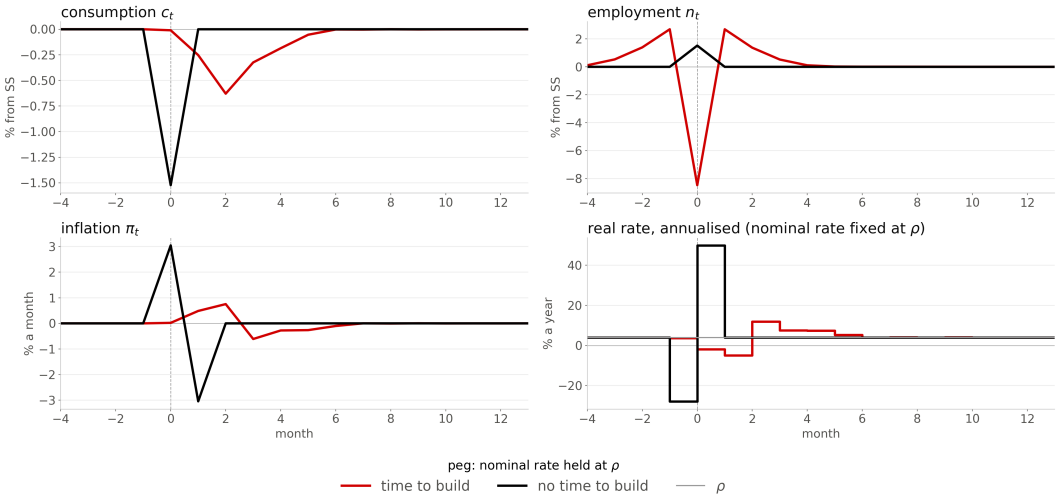
# Shock announced 3 periods ahead: firm profits



# Shock announced 3 periods ahead: welfare gain from optimal policy



# Shock announced 4 periods ahead: TFP shock, no policy change



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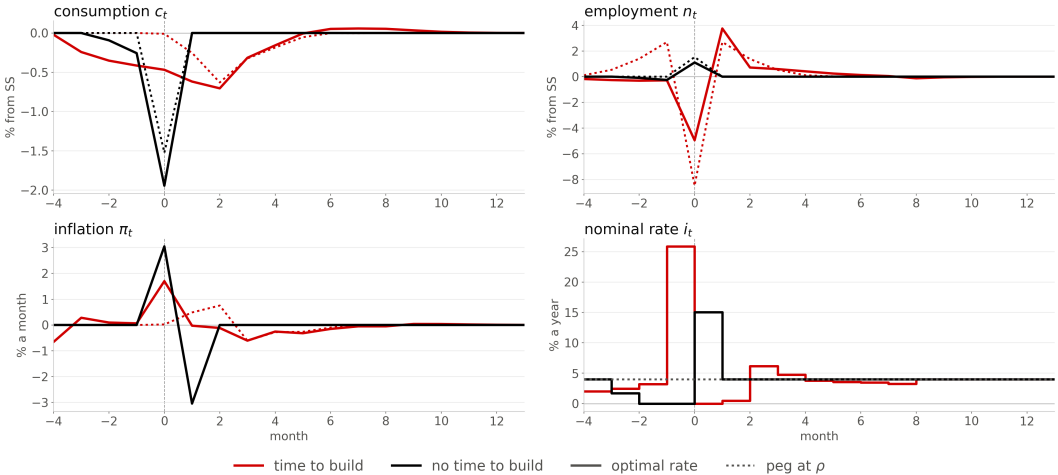
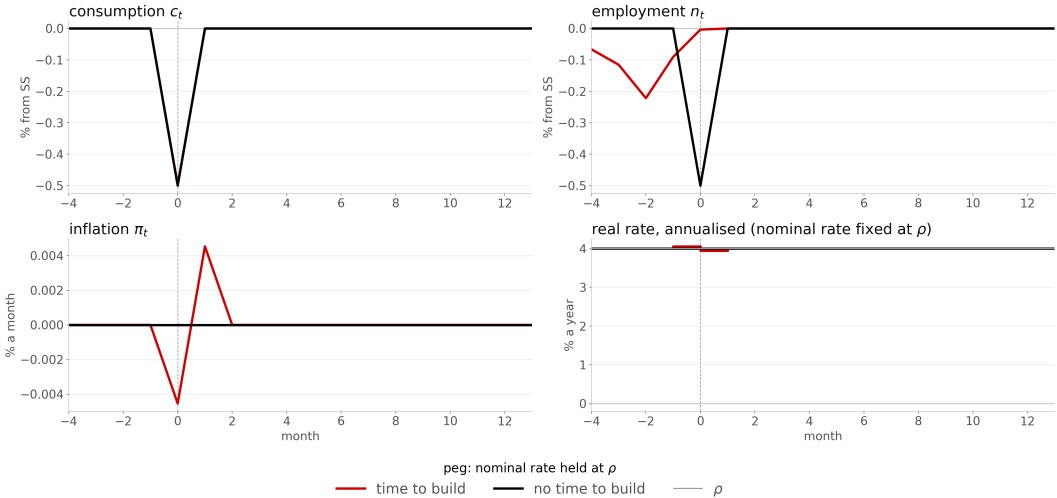


# Shock announced 4 periods ahead: demand shock, no policy change



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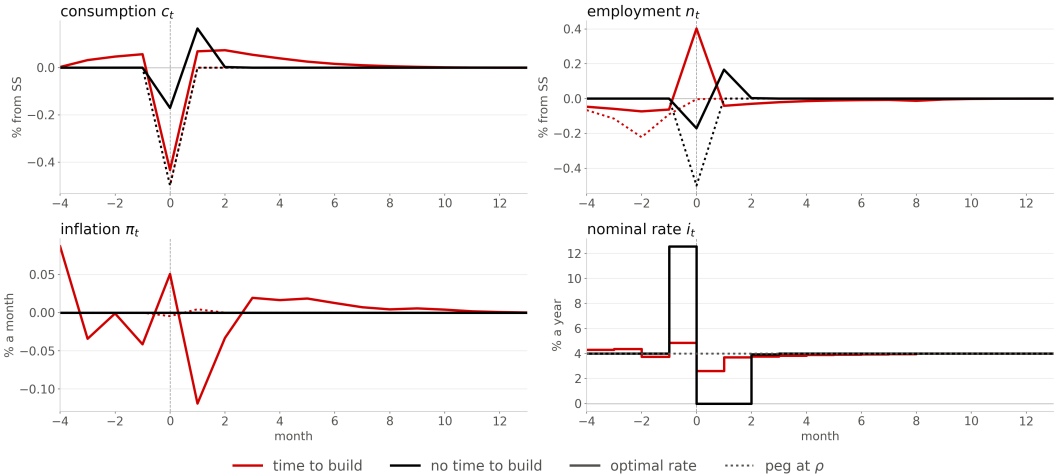
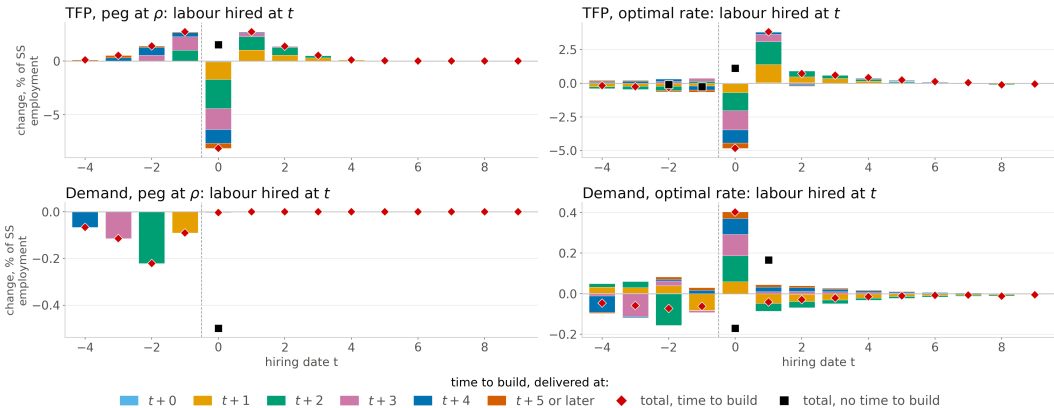
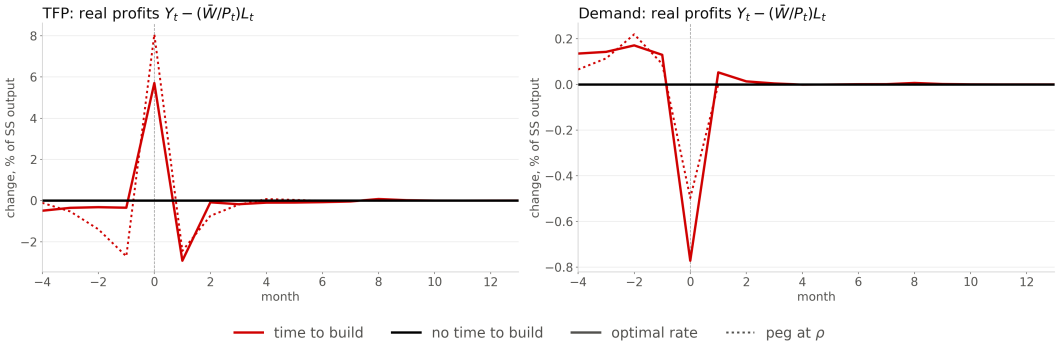


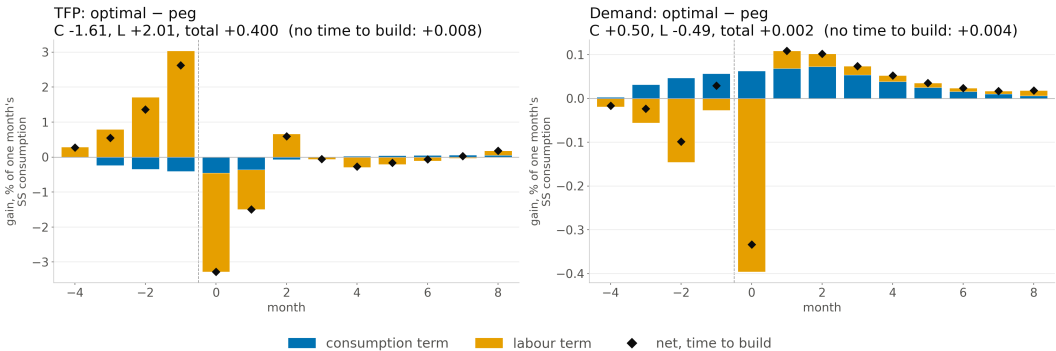
# Shock announced 4 periods ahead: where labour goes



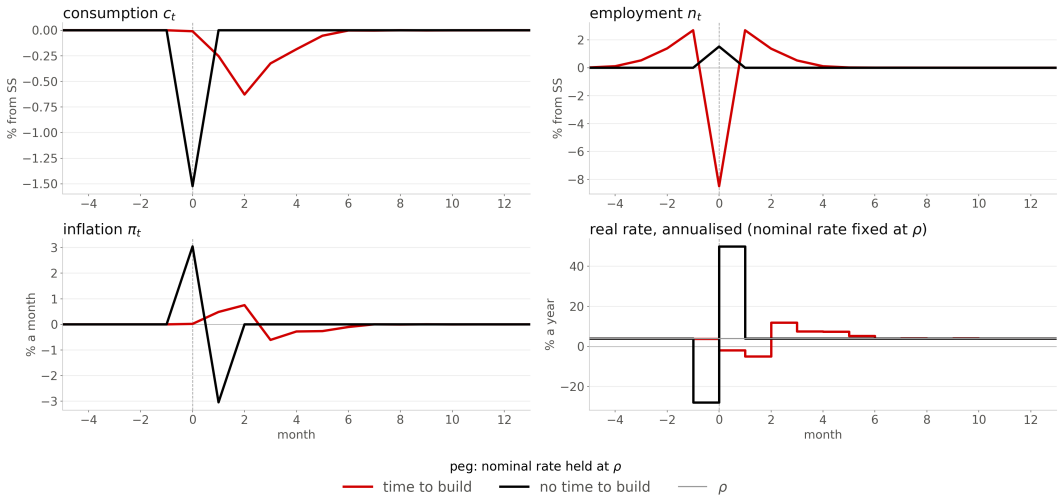
# Shock announced 4 periods ahead: firm profits



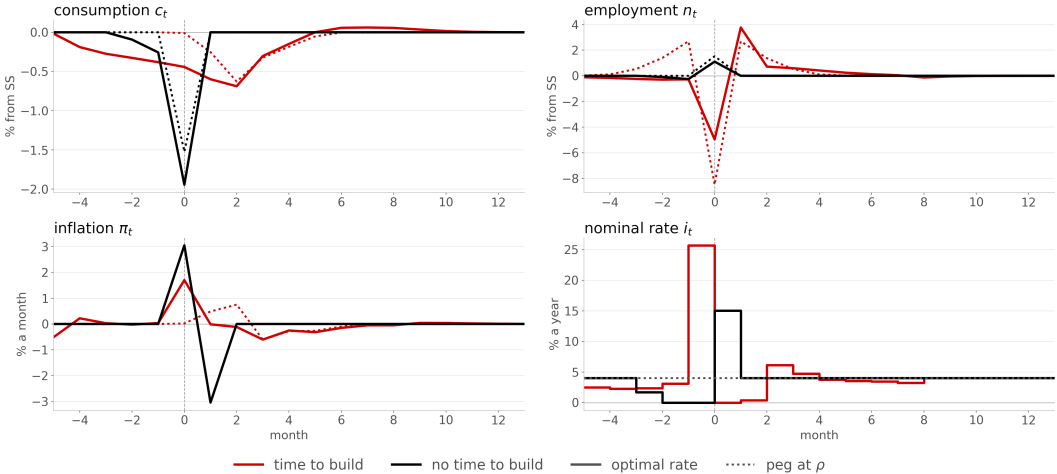
# Shock announced 4 periods ahead: welfare gain from optimal policy



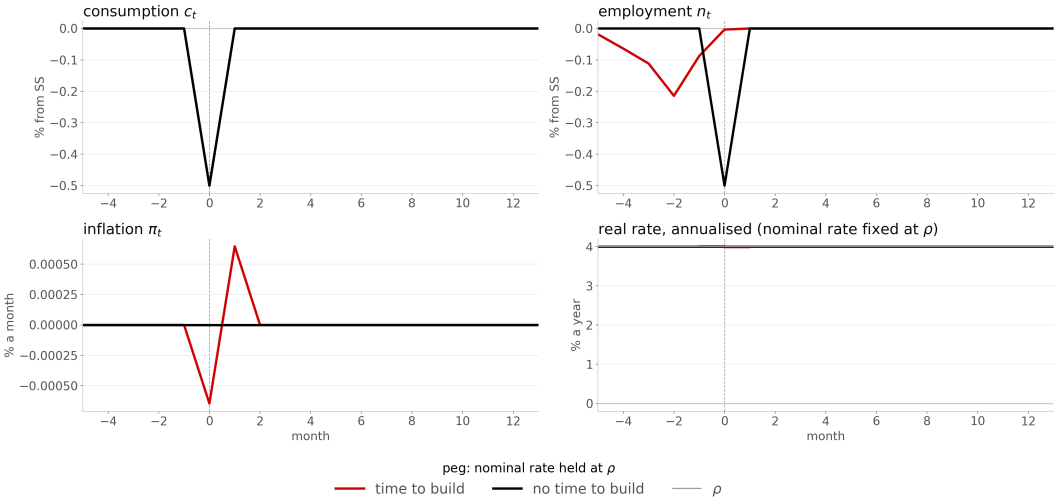
# Shock announced 5 periods ahead: TFP shock, no policy change



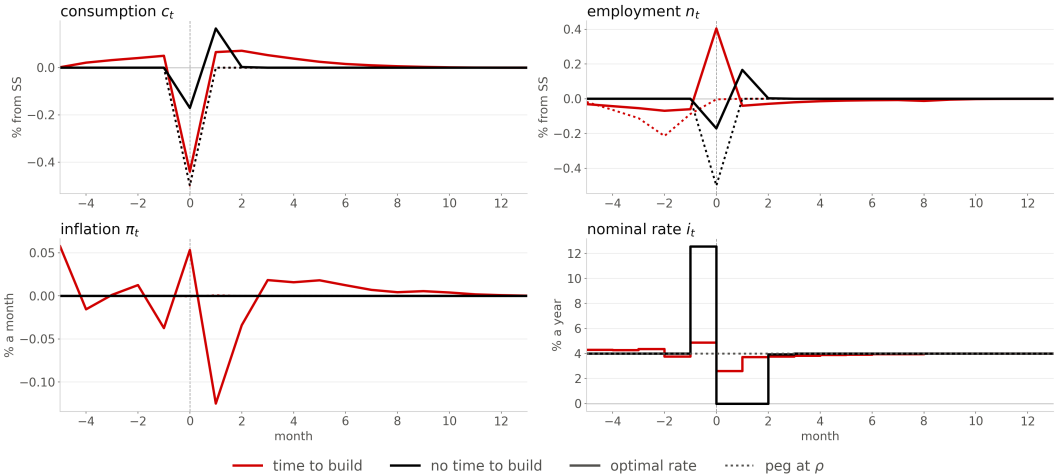
# Shock announced 5 periods ahead: TFP shock



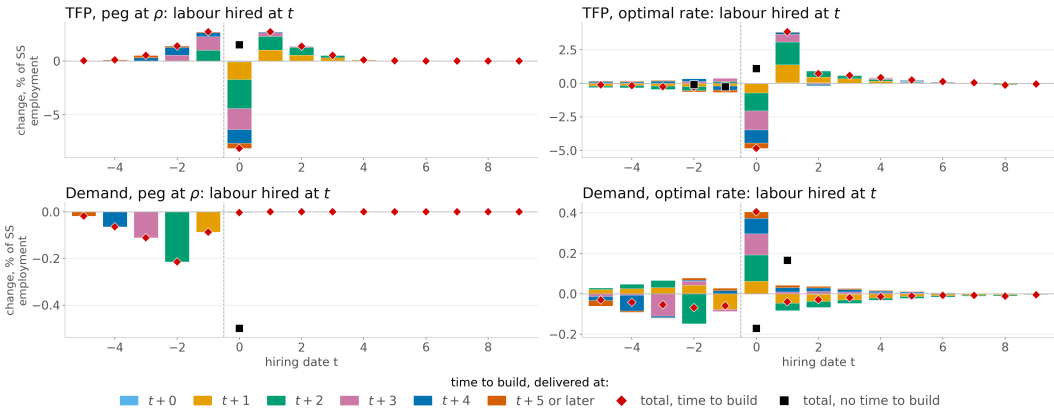
# Shock announced 5 periods ahead: demand shock, no policy change



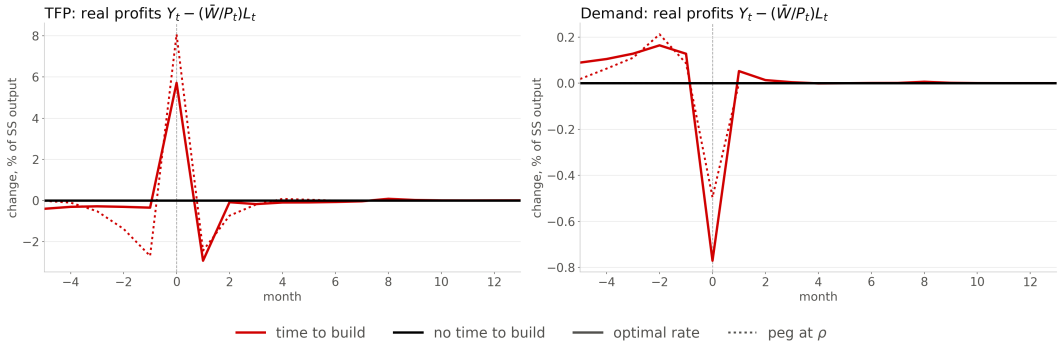
# Shock announced 5 periods ahead: demand shock



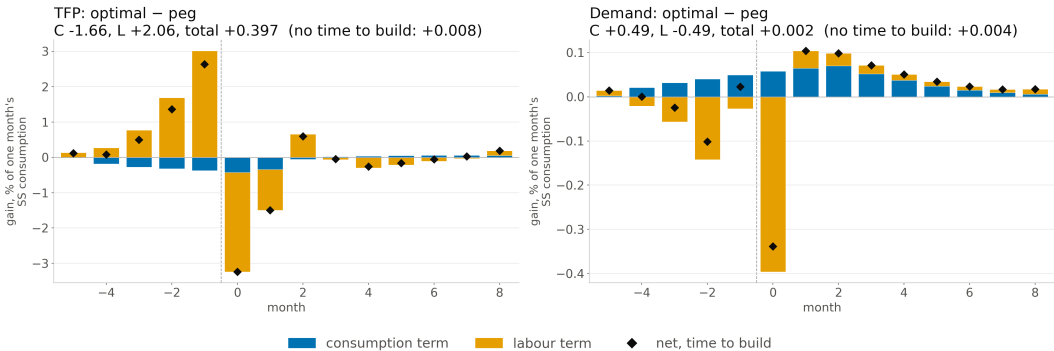
# Shock announced 5 periods ahead: where labour goes



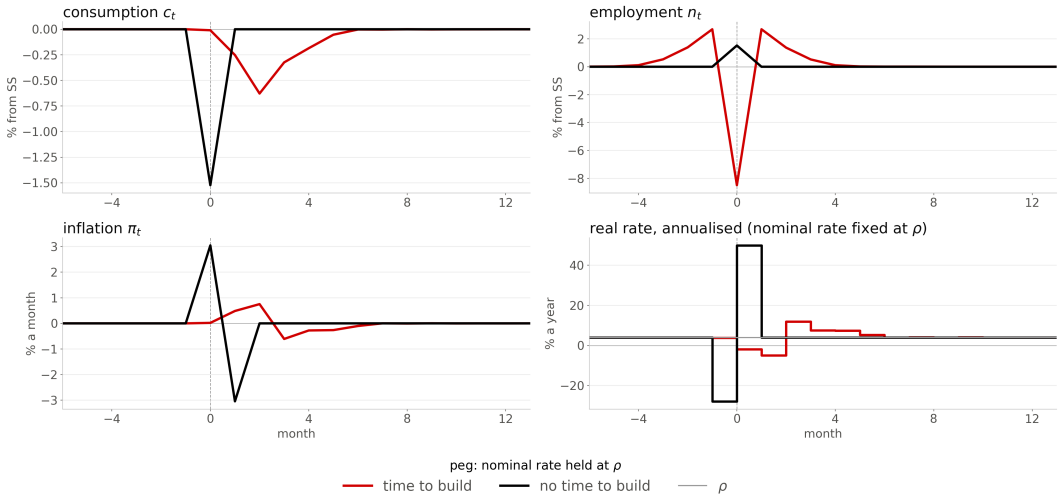
# Shock announced 5 periods ahead: firm profits



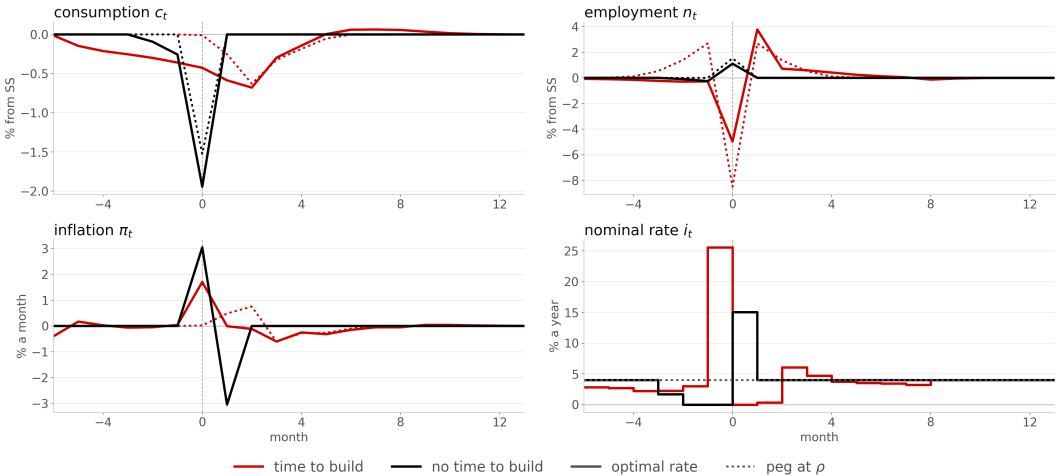
# Shock announced 5 periods ahead: welfare gain from optimal policy



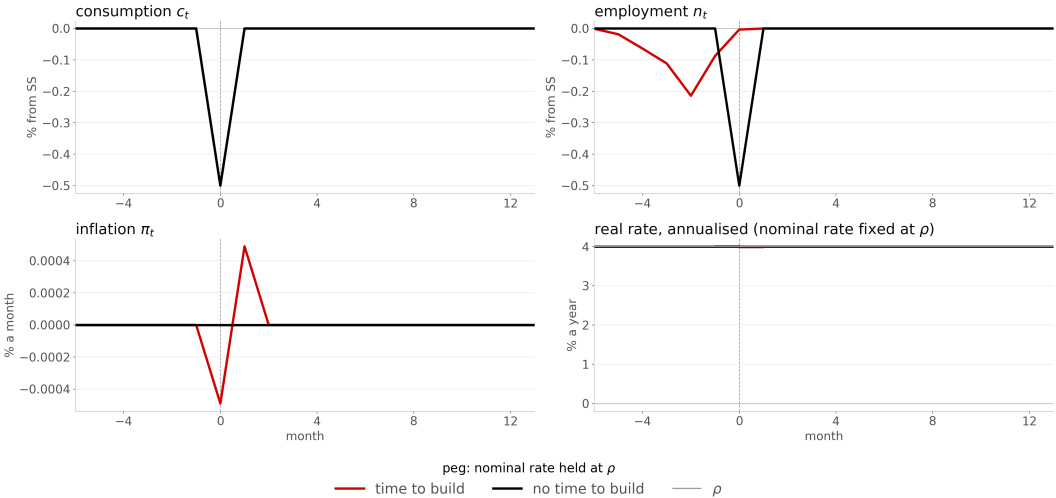
# Shock announced 6 periods ahead: TFP shock, no policy change



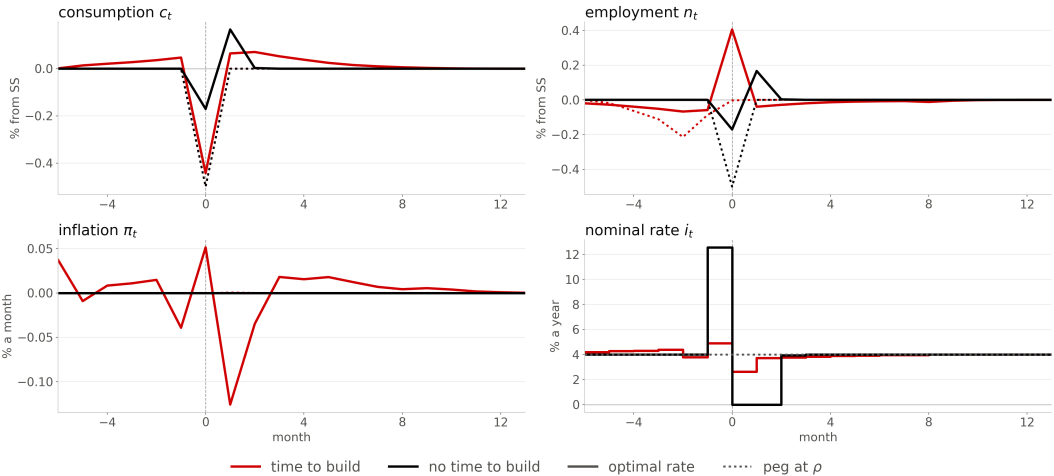
# Shock announced 6 periods ahead: TFP shock



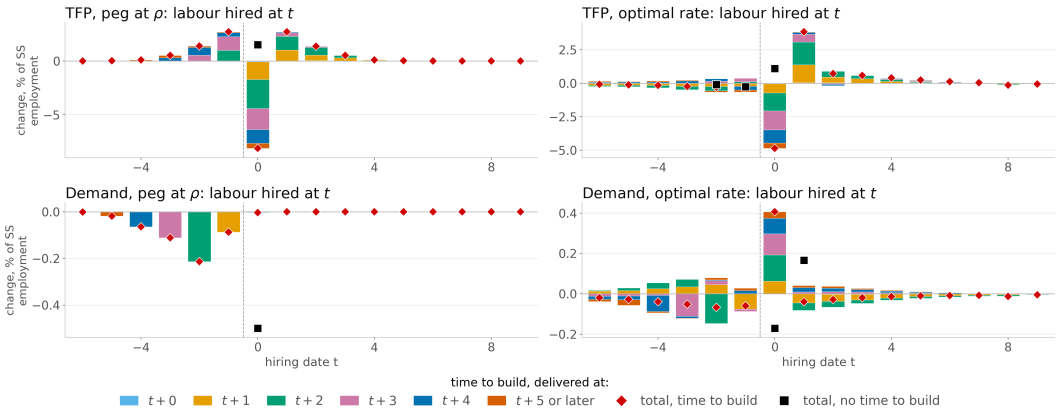
# Shock announced 6 periods ahead: demand shock, no policy change



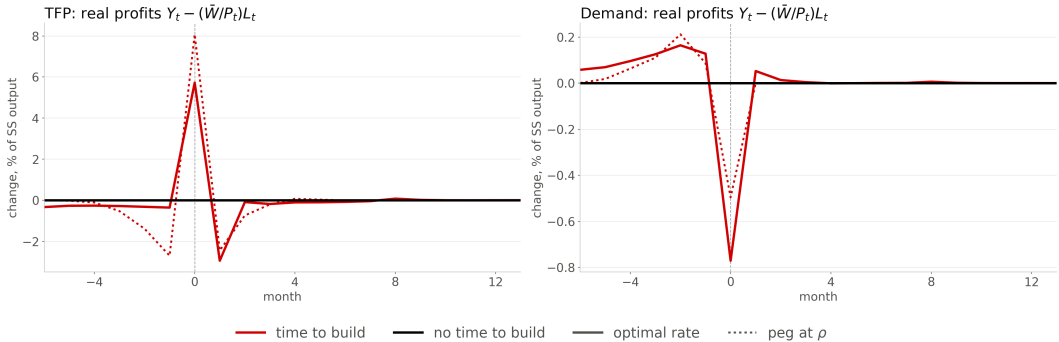
# Shock announced 6 periods ahead: demand shock



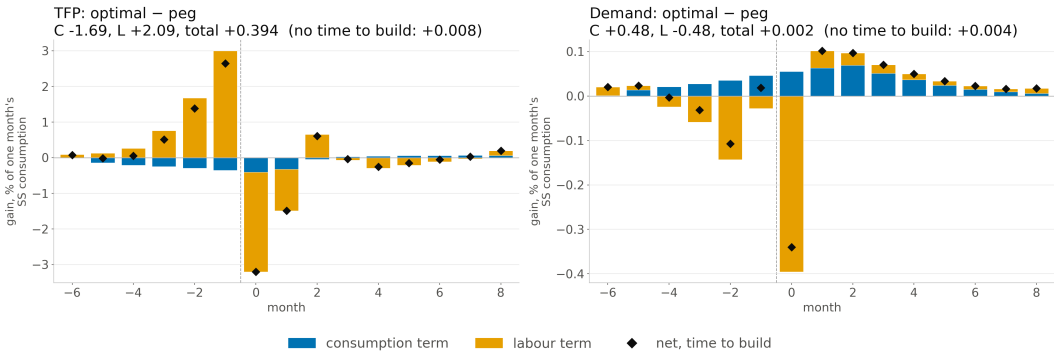
# Shock announced 6 periods ahead: where labour goes



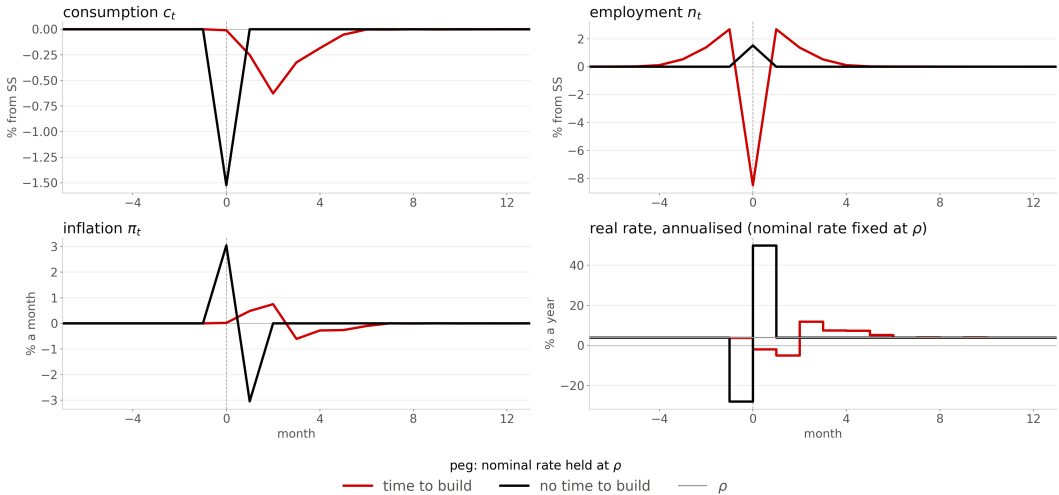
# Shock announced 6 periods ahead: firm profits



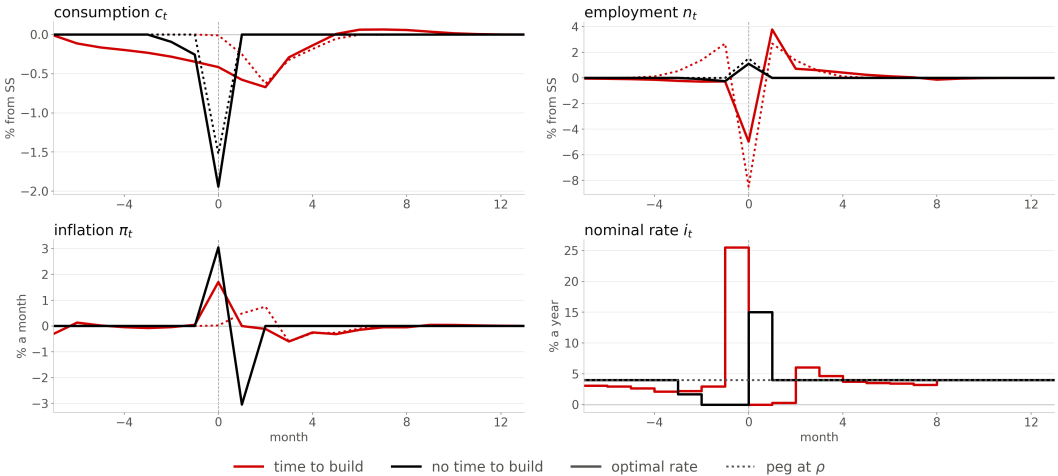
# Shock announced 6 periods ahead: welfare gain from optimal policy



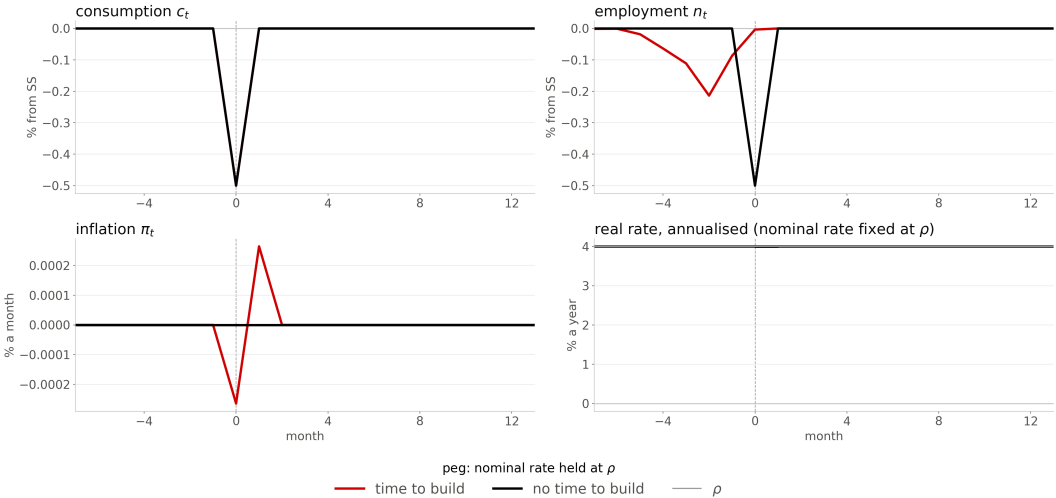
# Shock announced 7 periods ahead: TFP shock, no policy change



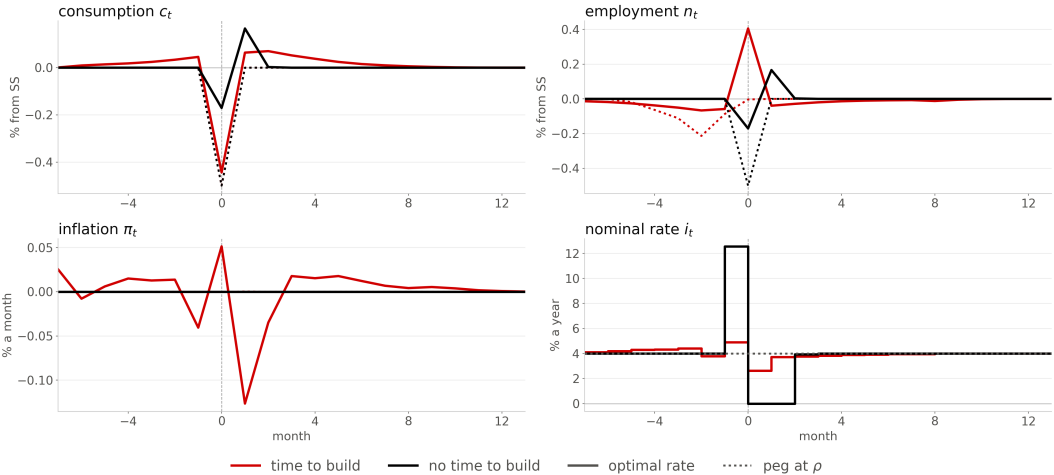
# Shock announced 7 periods ahead: TFP shock



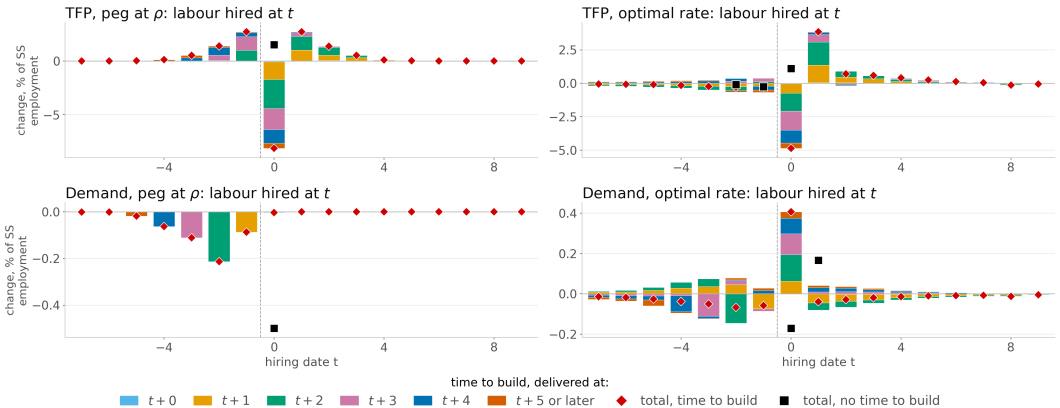
# Shock announced 7 periods ahead: demand shock, no policy change



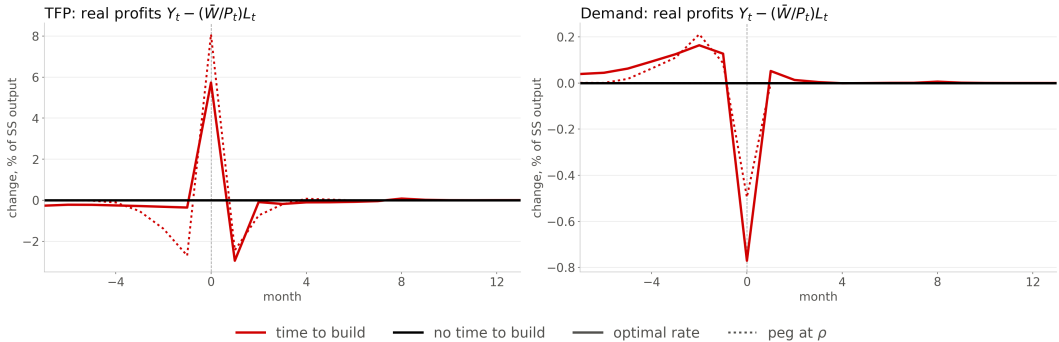
# Shock announced 7 periods ahead: demand shock



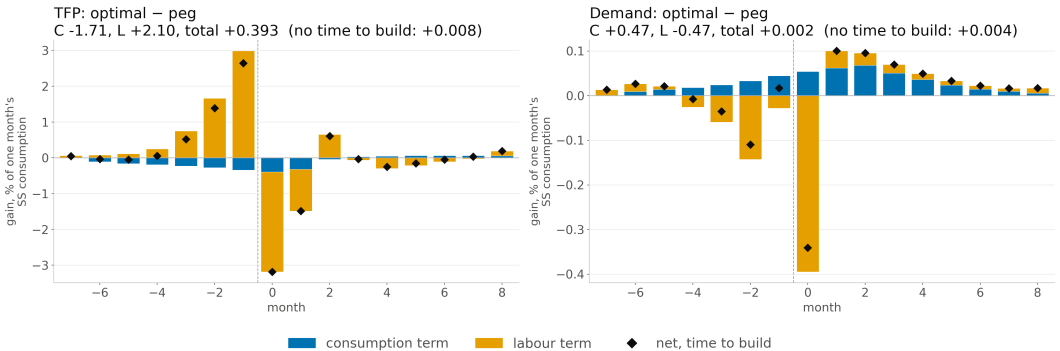
# Shock announced 7 periods ahead: where labour goes



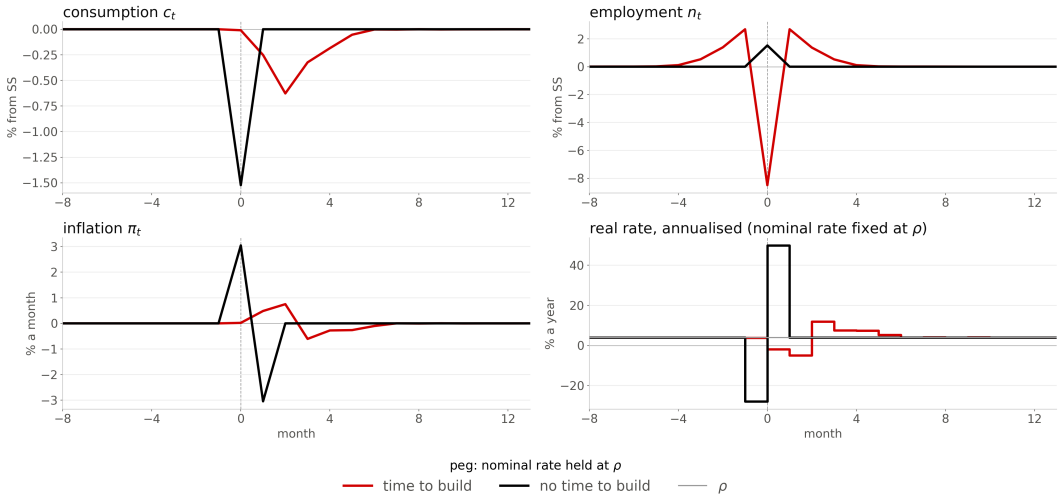
# Shock announced 7 periods ahead: firm profits



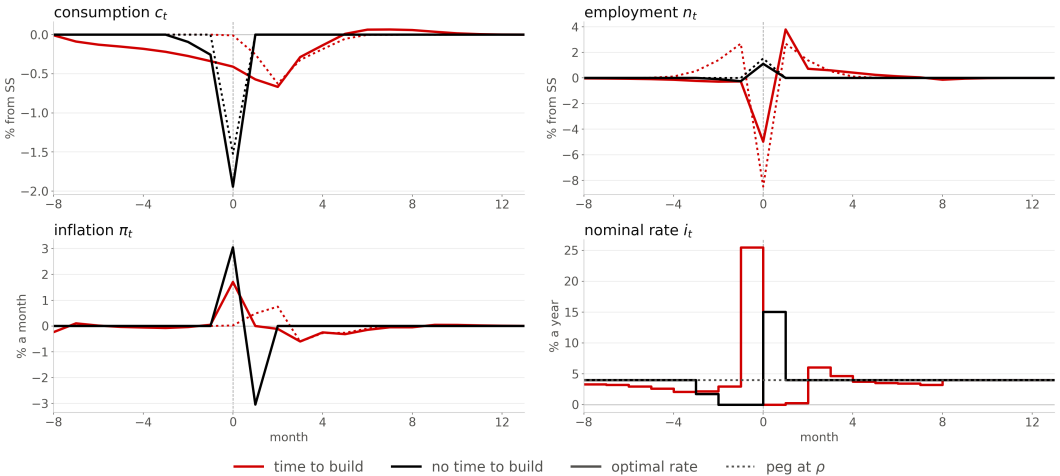
# Shock announced 7 periods ahead: welfare gain from optimal policy



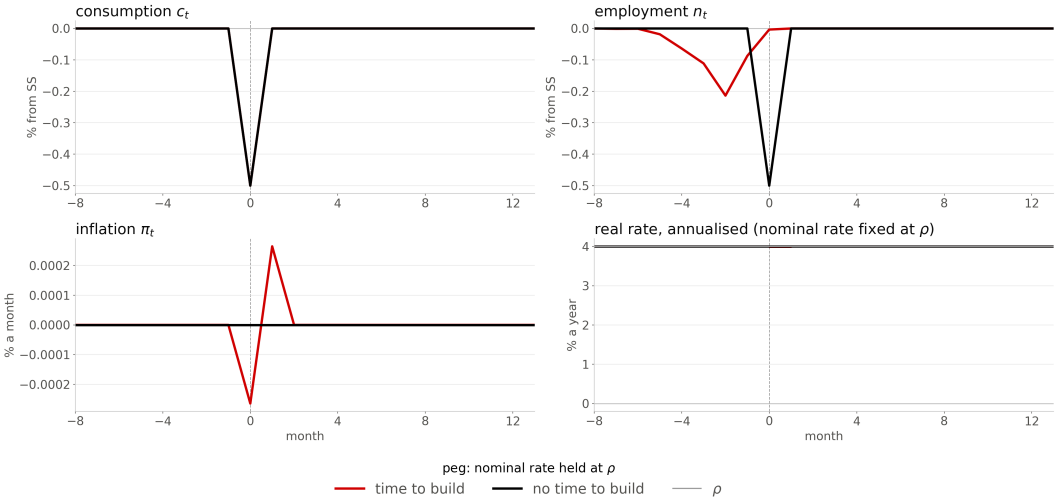
# Shock announced 8 periods ahead: TFP shock, no policy change



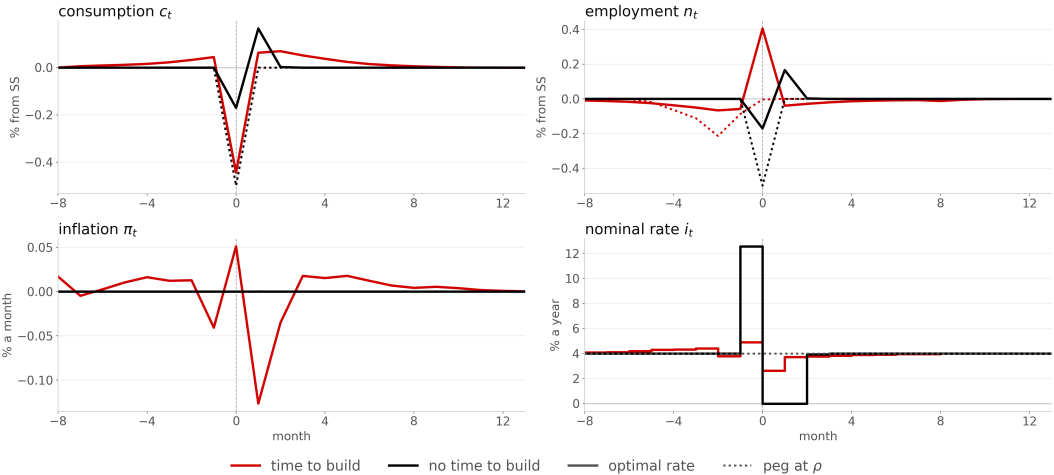
# Shock announced 8 periods ahead: TFP shock



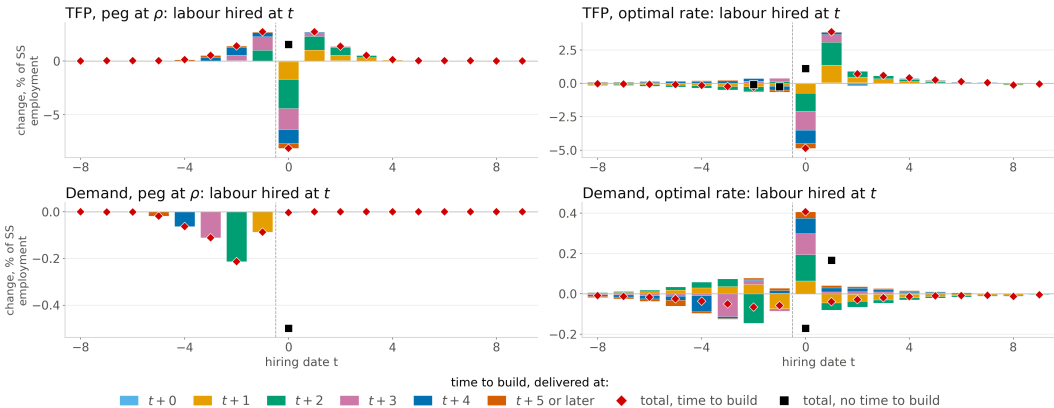
# Shock announced 8 periods ahead: demand shock, no policy change



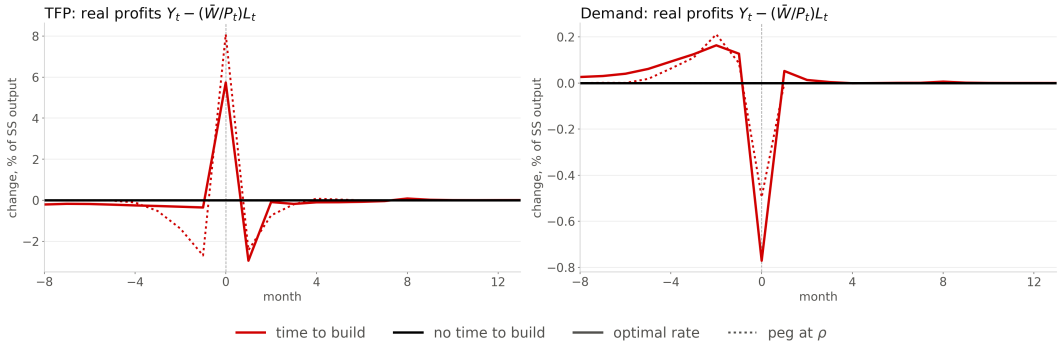
# Shock announced 8 periods ahead: demand shock



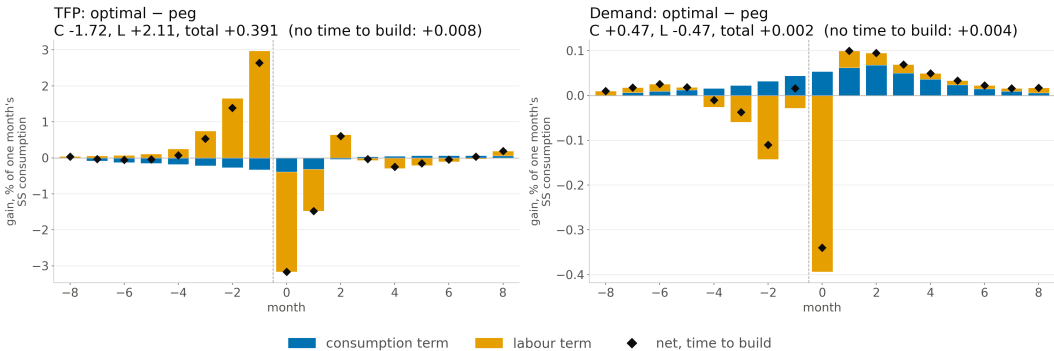
# Shock announced 8 periods ahead: where labour goes



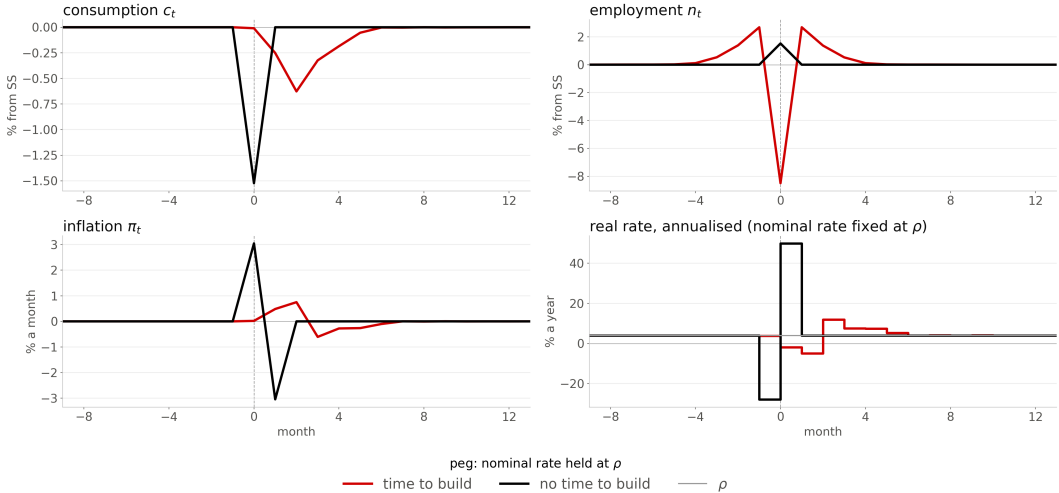
# Shock announced 8 periods ahead: firm profits



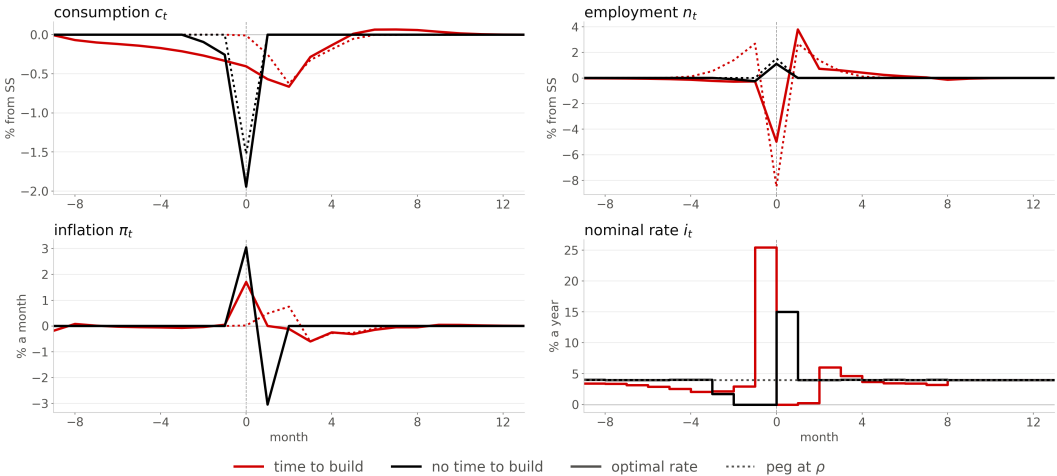
# Shock announced 8 periods ahead: welfare gain from optimal policy



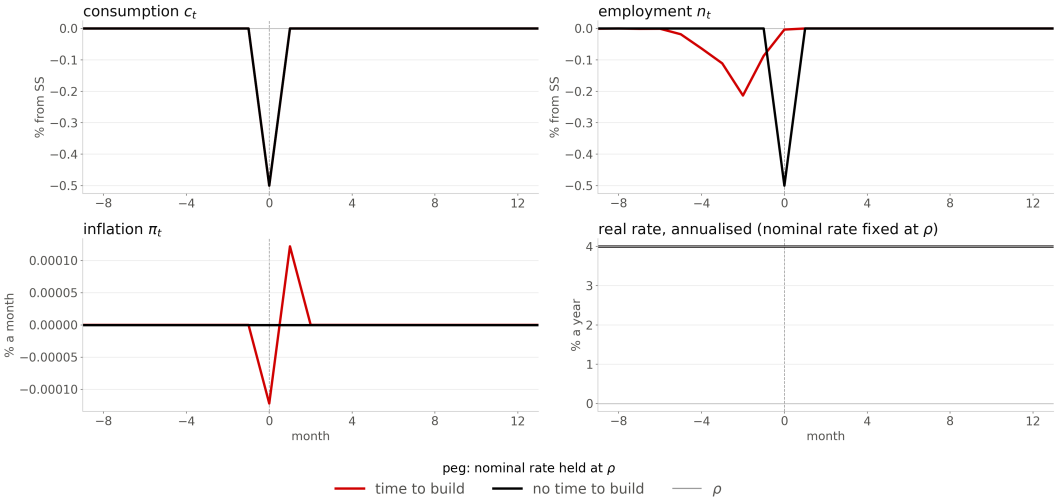
# Shock announced 9 periods ahead: TFP shock, no policy change



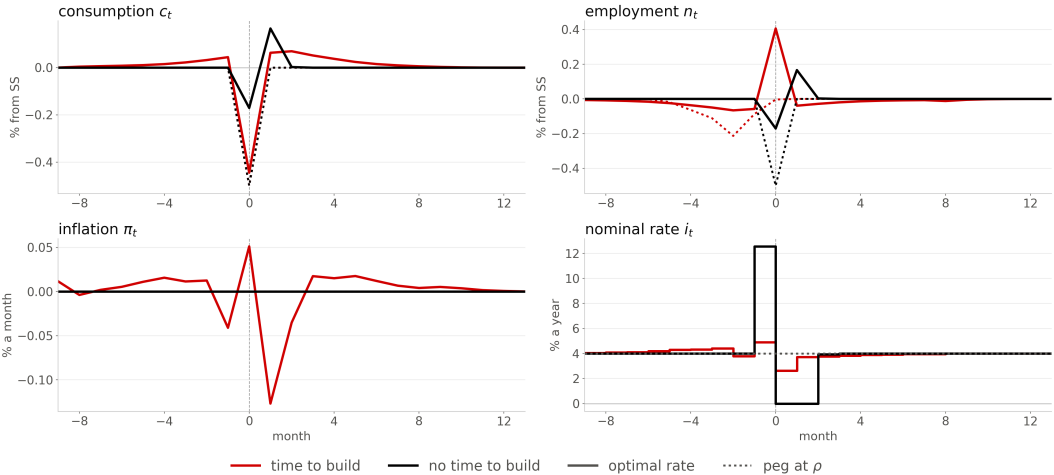
# Shock announced 9 periods ahead: TFP shock



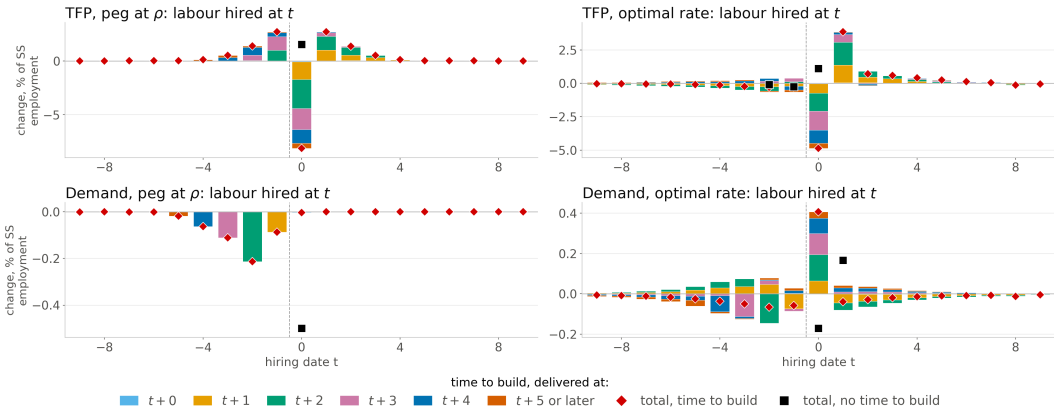
# Shock announced 9 periods ahead: demand shock, no policy change



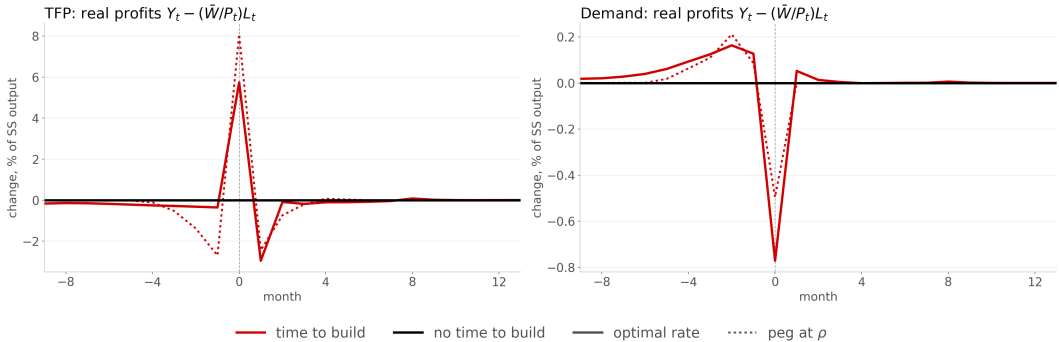
# Shock announced 9 periods ahead: demand shock



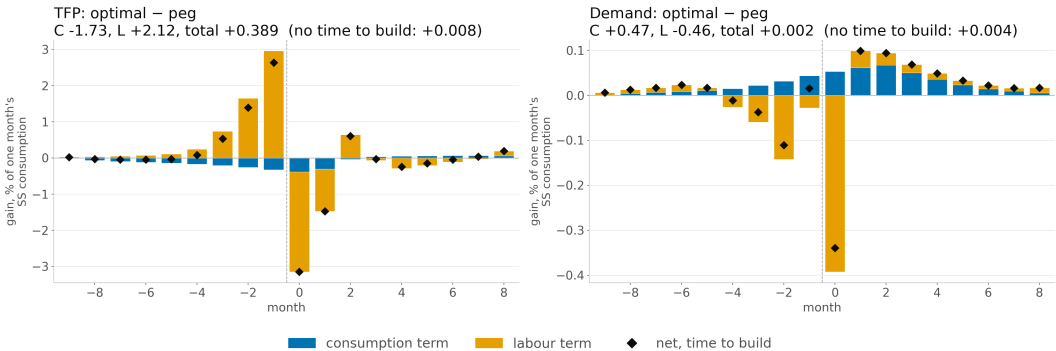
# Shock announced 9 periods ahead: where labour goes



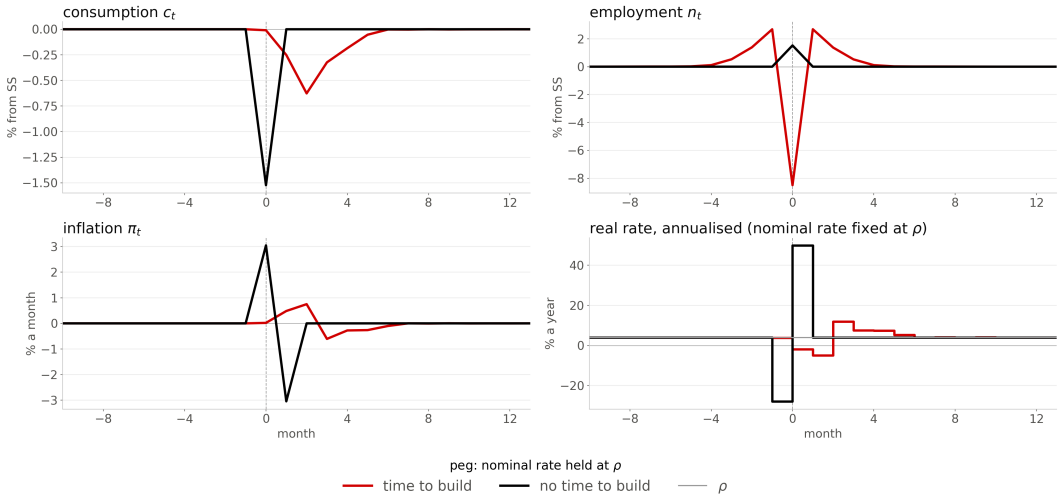
# Shock announced 9 periods ahead: firm profits



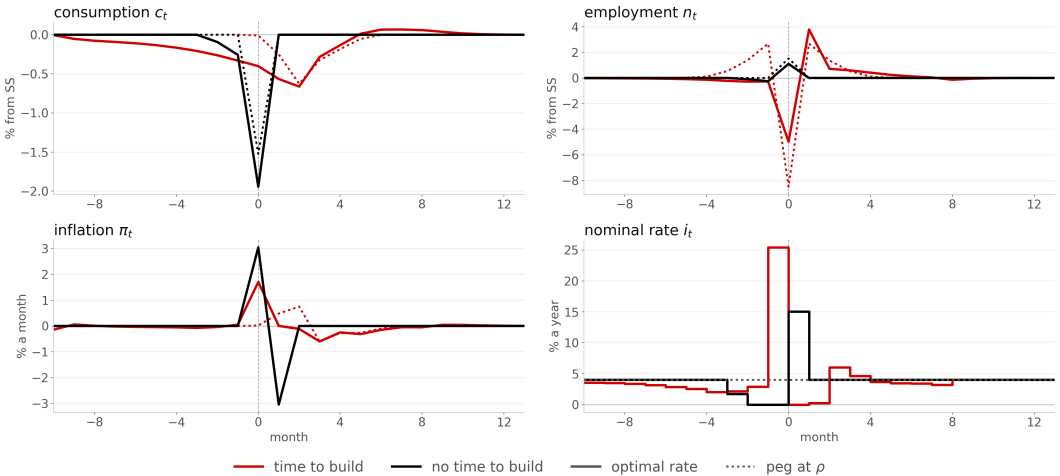
# Shock announced 9 periods ahead: welfare gain from optimal policy



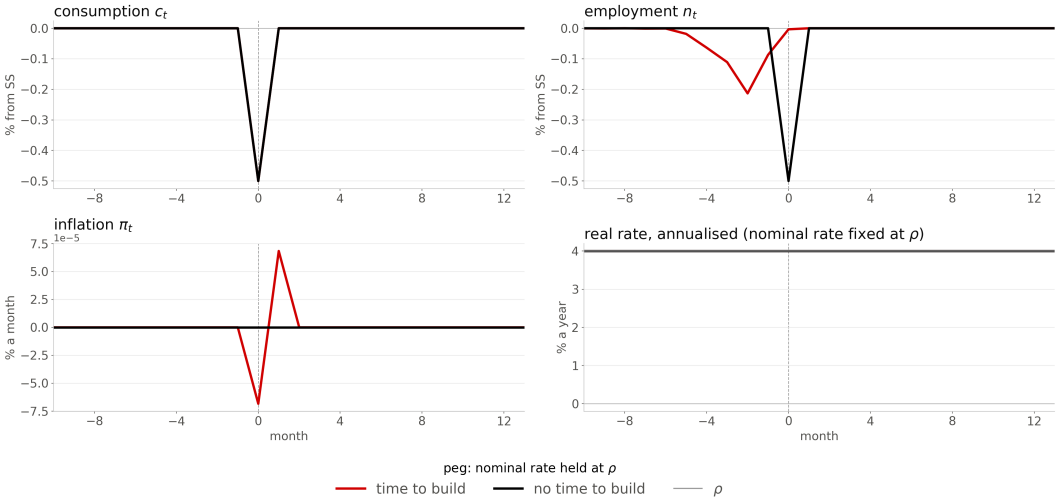
# Shock announced 10 periods ahead: TFP shock, no policy change



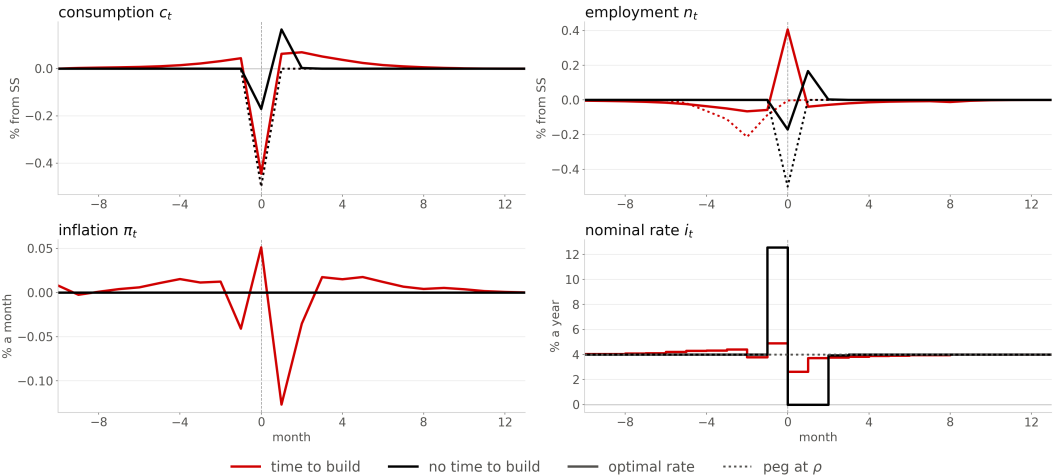
# Shock announced 10 periods ahead: TFP shock



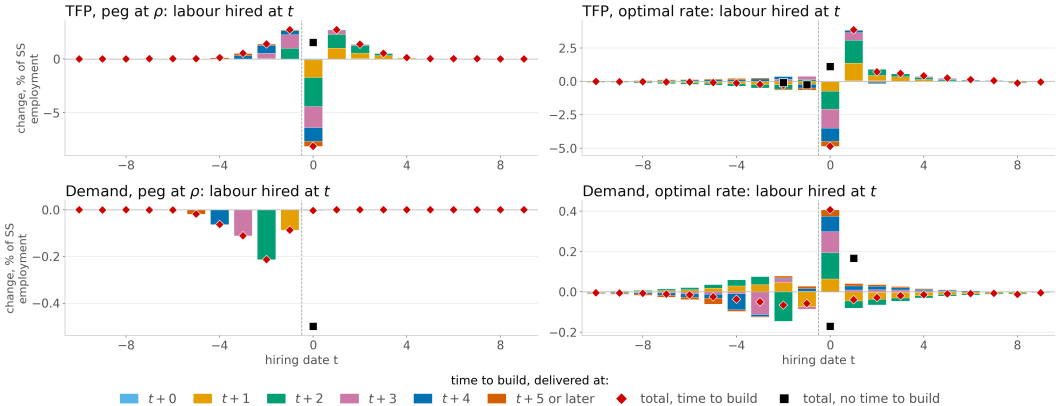
# Shock announced 10 periods ahead: demand shock, no policy change



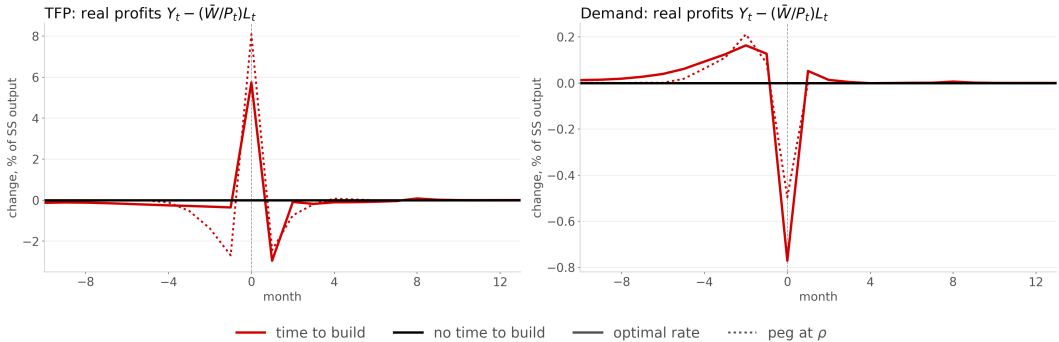
# Shock announced 10 periods ahead: demand shock



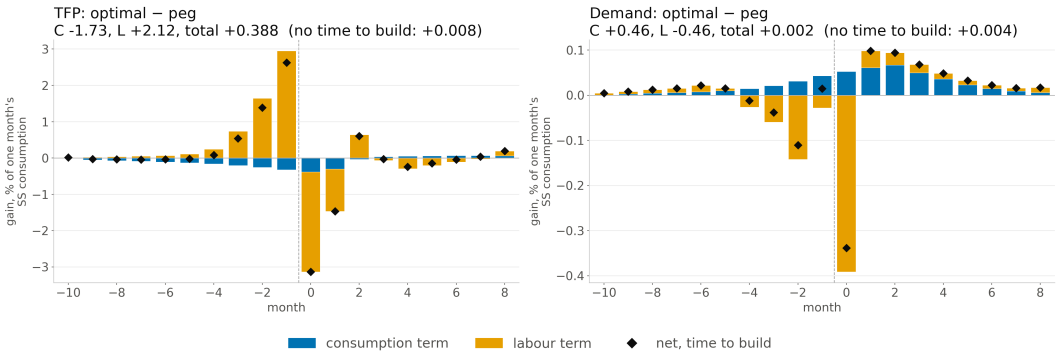
# Shock announced 10 periods ahead: where labour goes



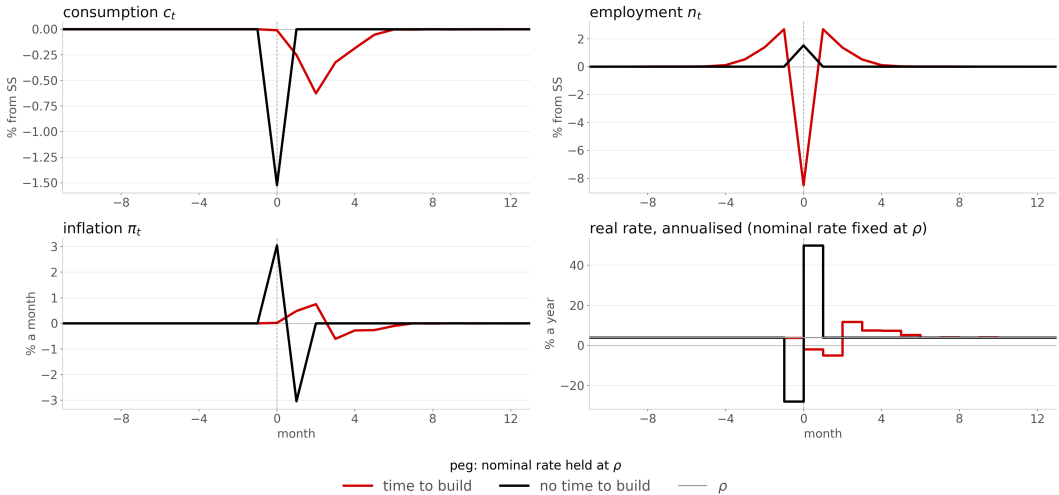
# Shock announced 10 periods ahead: firm profits



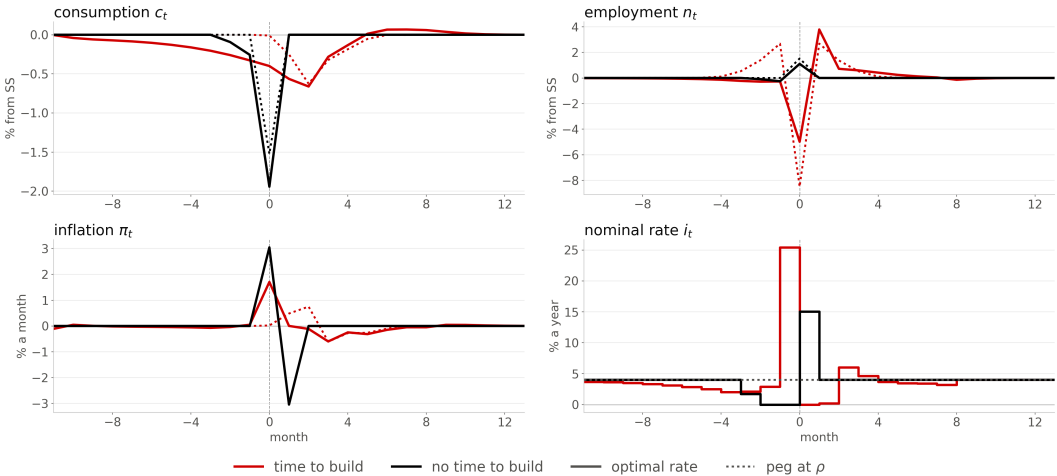
# Shock announced 10 periods ahead: welfare gain from optimal policy



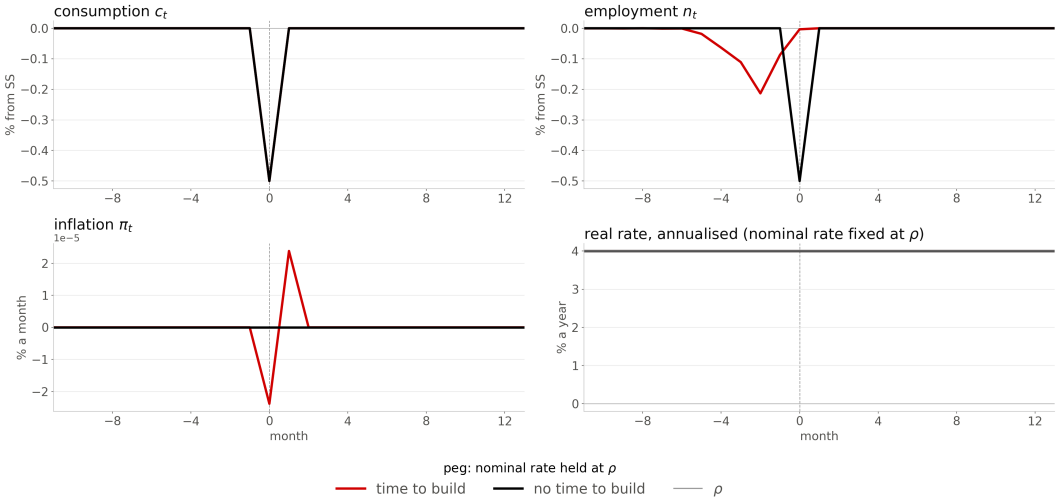
# Shock announced 11 periods ahead: TFP shock, no policy change



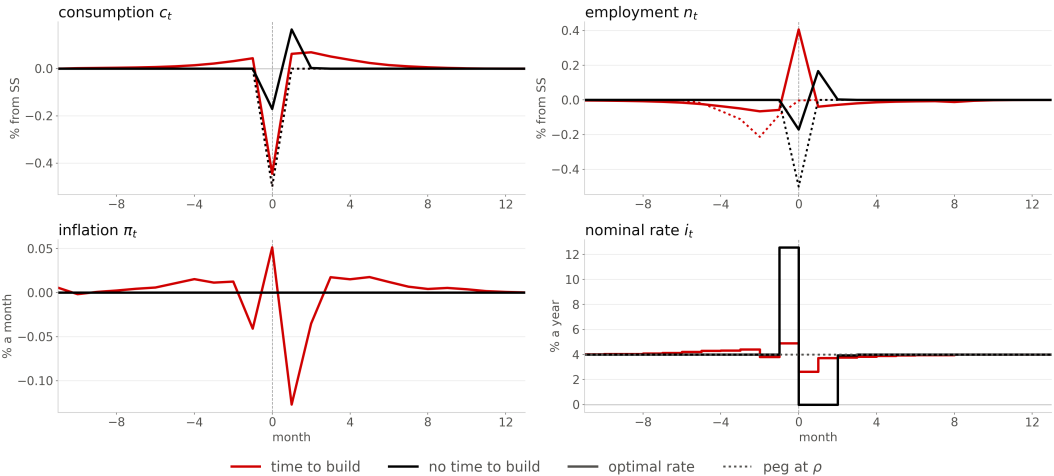
# Shock announced 11 periods ahead: TFP shock



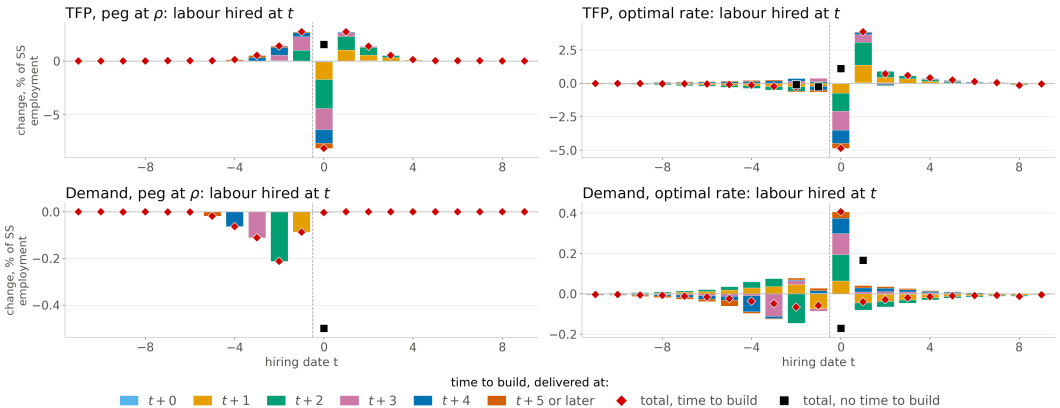
# Shock announced 11 periods ahead: demand shock, no policy change



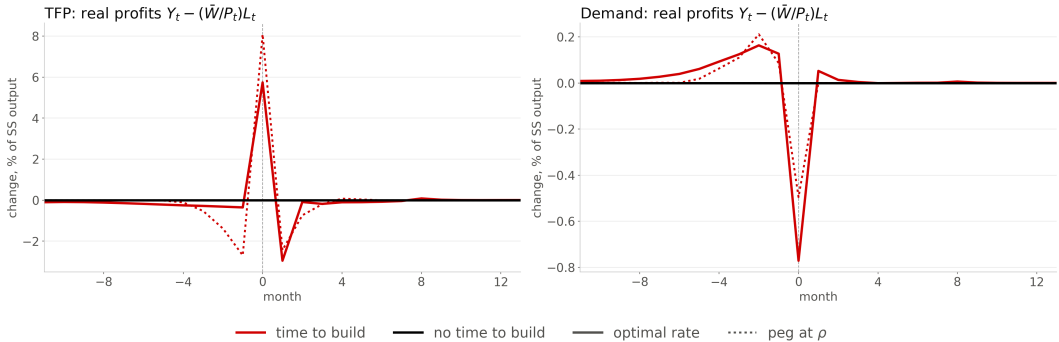
# Shock announced 11 periods ahead: demand shock



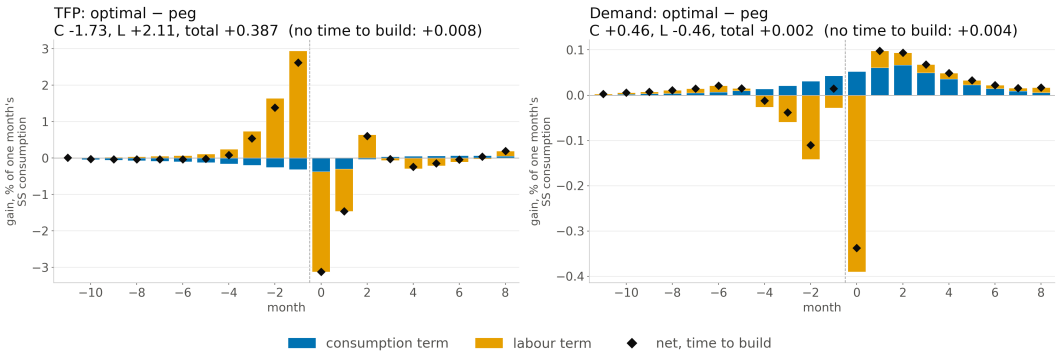
# Shock announced 11 periods ahead: where labour goes



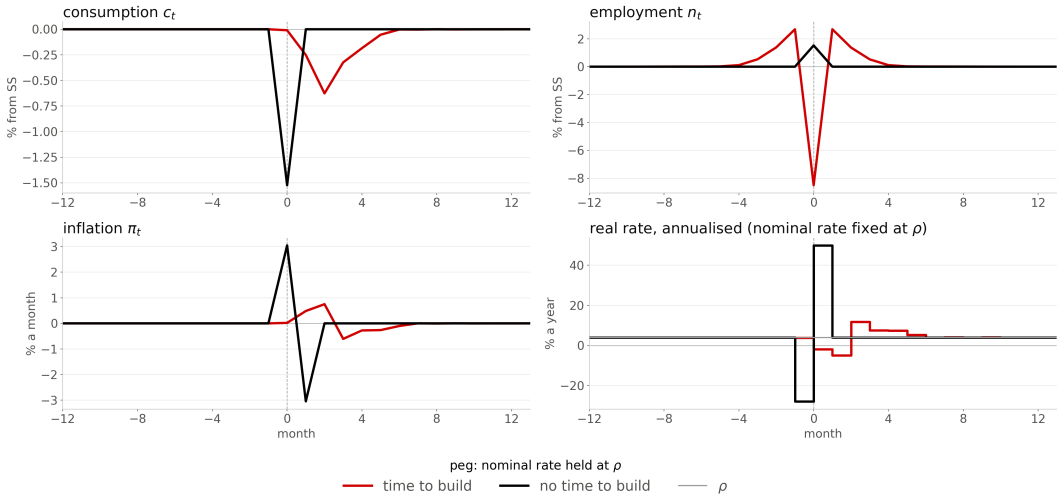
# Shock announced 11 periods ahead: firm profits



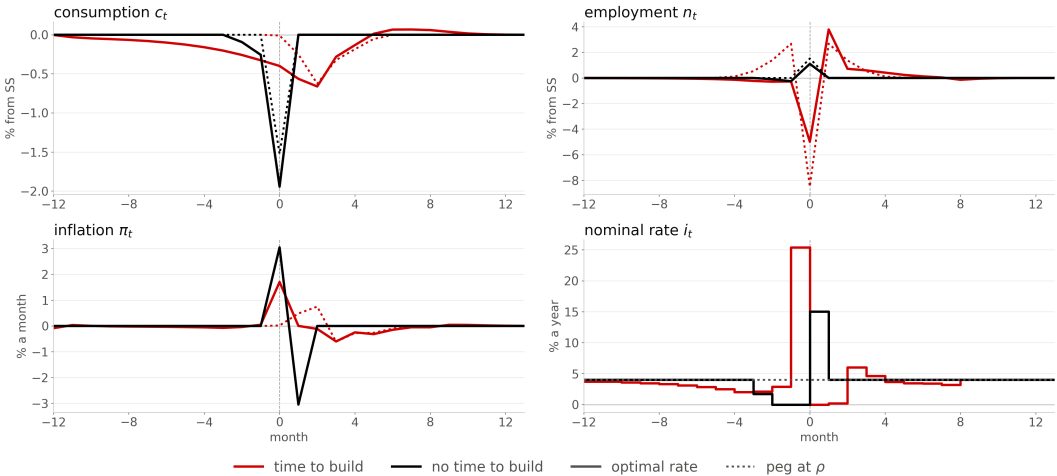
# Shock announced 11 periods ahead: welfare gain from optimal policy



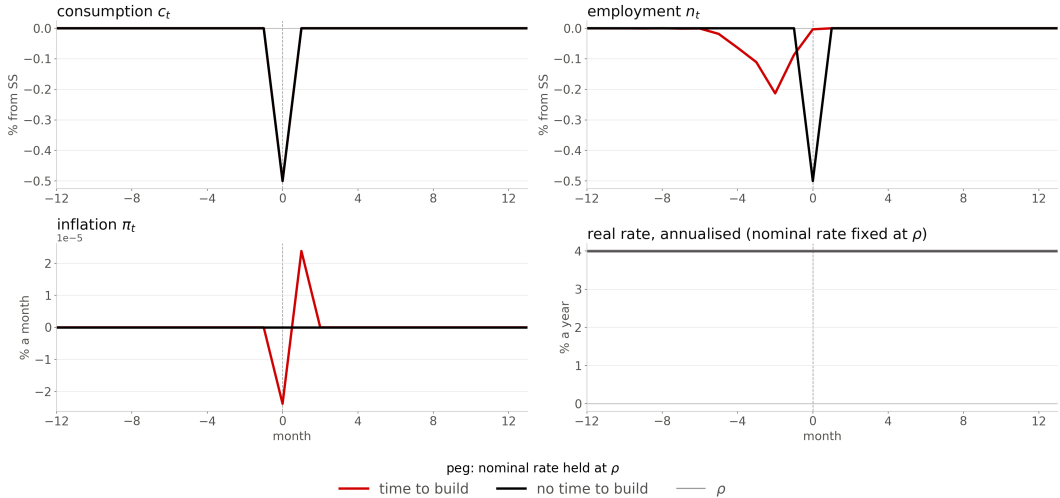
# Shock announced 12 periods ahead: TFP shock, no policy change



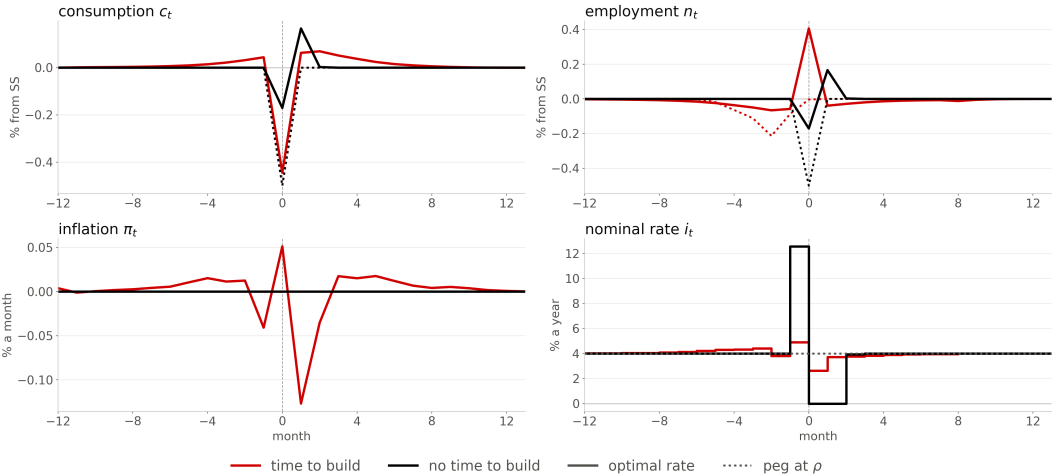
# Shock announced 12 periods ahead: TFP shock



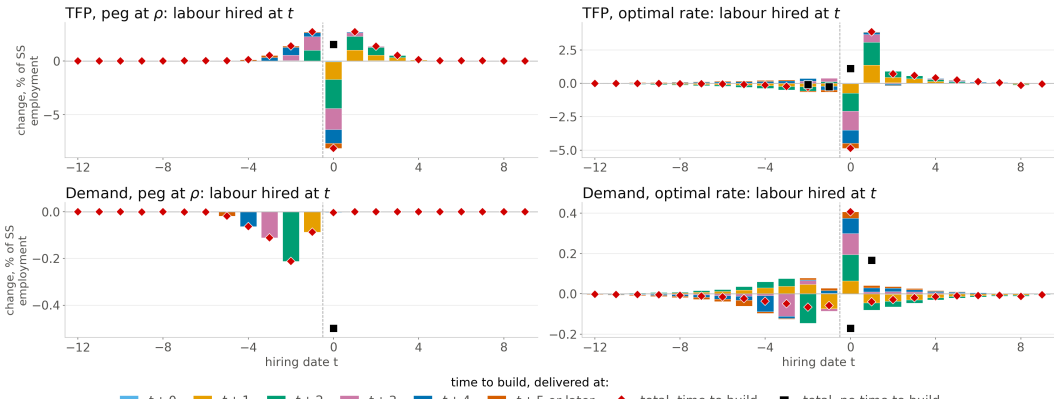
# Shock announced 12 periods ahead: demand shock, no policy change



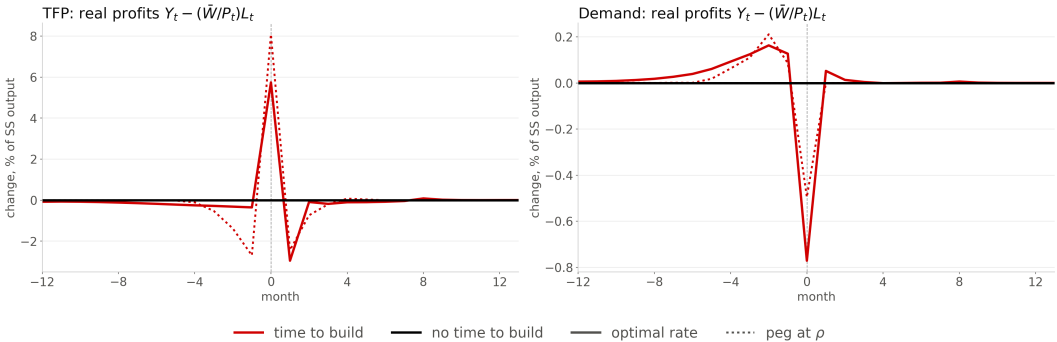
# Shock announced 12 periods ahead: demand shock



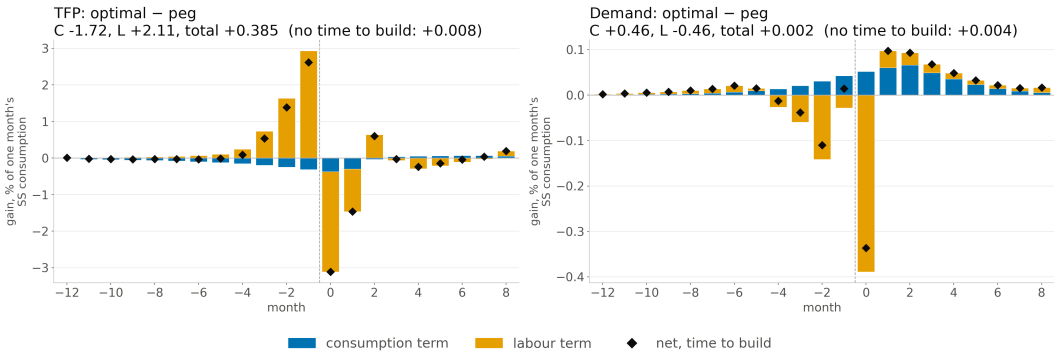
# Shock announced 12 periods ahead: where labour goes



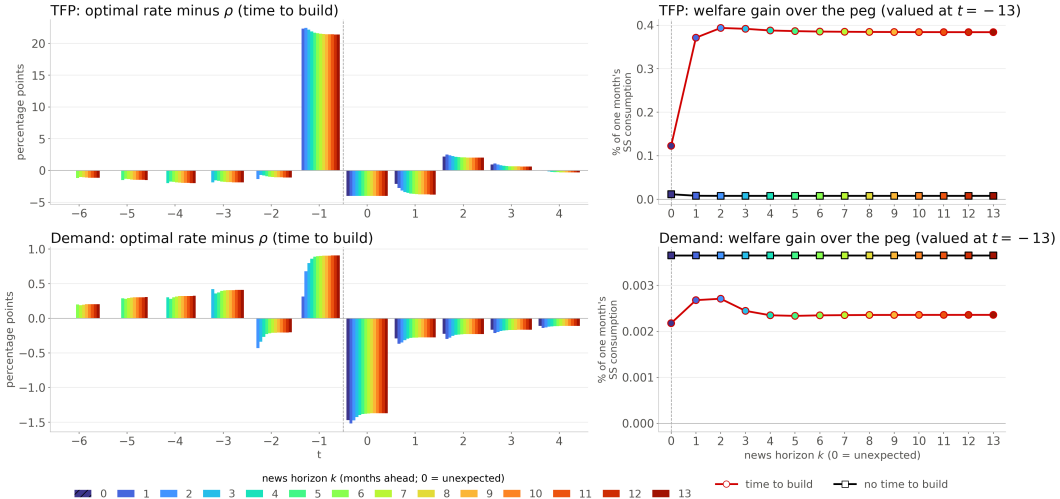
# Shock announced 12 periods ahead: firm profits



# Shock announced 12 periods ahead: welfare gain from optimal policy



# News horizons: summary

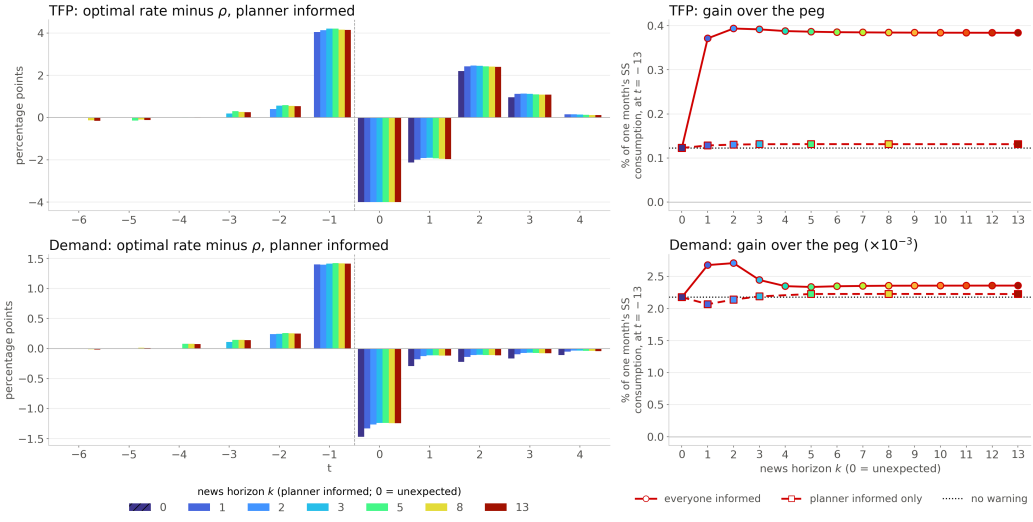


# News horizons: welfare gain from optimal policy

| $k$ | TFP | demand ($\times 10^{-3}$) |
|---|---|---|
| 0 (unexp.) | 0.123 | 2.18 |
| 1 | 0.371 | 2.68 |
| 2 | 0.394 | 2.71 |
| 3 | 0.392 | 2.45 |
| 4 | 0.388 | 2.35 |
| 5 | 0.387 | 2.34 |
| 6 | 0.385 | 2.35 |
| 7 | 0.385 | 2.35 |
| 8 | 0.385 | 2.36 |
| 9 | 0.384 | 2.36 |
| 10 | 0.384 | 2.36 |
| 11 | 0.384 | 2.36 |
| 12 | 0.384 | 2.36 |
| 13 | 0.384 | 2.36 |

Time to build. Gain = optimal − peg, % of one month's SS consumption, valued at $t = -13$. No time to build: 0.011 / 0.008 (TFP), 3.65 (demand).

# News horizons: planner informed, firms and households not



# Who should know? Household welfare under the optimal rate

| | TFP | | | demand ($\times 10^{-3}$) | | |
|---|---|---|---|---|---|---|
| $k$ | planner only | everyone | difference | planner only | everyone | difference |
| 0 (no warning) | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | $+0.006$ | $-0.029$ | **$+0.035$** | $-0.11$ | $+1.13$ | **$-1.24$** |
| 2 | $+0.008$ | $-0.041$ | **$+0.048$** | $-0.04$ | $+1.66$ | **$-1.70$** |
| 3 | $+0.009$ | $-0.047$ | **$+0.055$** | $+0.01$ | $+1.85$ | **$-1.84$** |
| 5 | $+0.009$ | $-0.051$ | **$+0.060$** | $+0.05$ | $+1.99$ | **$-1.94$** |
| 8 | $+0.009$ | $-0.053$ | **$+0.062$** | $+0.05$ | $+2.02$ | **$-1.97$** |
| 13 | $+0.009$ | $-0.054$ | **$+0.063$** | $+0.05$ | $+2.02$ | **$-1.97$** |

Time to build. Utility under the optimal rate relative to no warning, % of one month's SS consumption, valued at $t = -13$. Difference = planner only $-$ everyone informed.

## Appendix: shock figures

- $c_t,\ n_t$: $100 \log(X_t / X^{ss})$; $n_t$ = labour **hired** at $t$
- $\pi_t = 100\,(\log P_t - \log P_{t-1})$ (% a month), $P_{-1} = P^{ss}$; $i_t$ annualised
- Both: exact Euler, $W_t = W^{ss}$, labour dated at hiring in $U$
- Expected: shock announced at $t = -13$ (months); unexpected: news at $t = 0$
- Optimal: $\max U$ over $i_t$ up to $t = 7$, then $\rho$; $i_t \in [0, 5\%]$
  free from $t = 0$, or from the news date ($t = -13$, expected only)
- No time to build: same firms, $\tau_j \equiv 0$

## Appendix: where labour goes

Labour hired at $t$ by firm $j$, for its delivery at $t + \tau_j$:

$$l_{j,t} = \frac{y_{j,t+\tau_j}}{\zeta_j A_t}$$

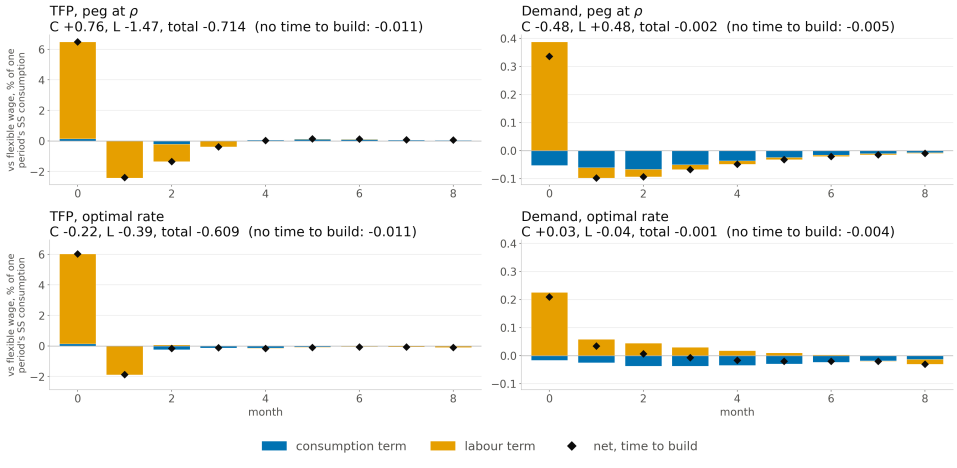
Bar for lag group $g \in \{0, 1, 2, 3, 4, 5^+\}$:

$$b_{g,t} = 100\,\frac{\sum_{j\in g} l_{j,t} - \sum_{j\in g} l_j^{ss}}{L^{ss}}, \qquad L^{ss} = \sum_j l_j^{ss}, \qquad \sum_g b_{g,t} = 100\,\frac{L_t - L^{ss}}{L^{ss}} \approx n_t$$

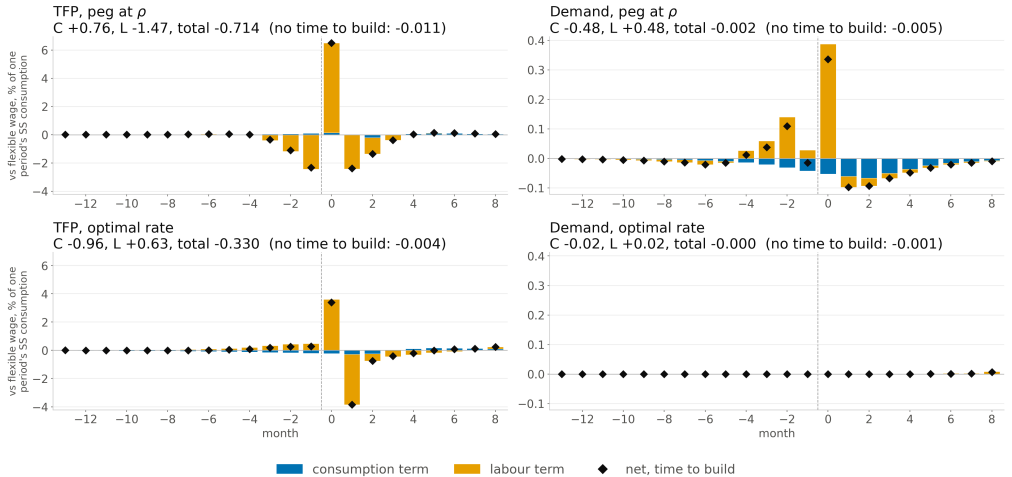
| $\tau$ | 0 | 1 | 2 | 3 | 4 | 5+ |
|---|---|---|---|---|---|---|
| SS share of $L$, % | 0.7 | 17.4 | 42.7 | 22.2 | 12.7 | 4.3 |

$b_{g,t}$ = (group's own % change) $\times$ (its share). Time to build only.

# Appendix: loss vs flexible wage, expected, policy from $t = 0$



# Appendix: loss vs flexible wage, expected, policy from news



# Appendix: loss vs flexible wage, unexpected